# Scale Drawings

*Teacher's Edition*

## The Period

**MA.7.GR.1.5** Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

**Must** Find the scale factor between two figures (k = copy ÷ original, in either direction of size) and use a scale factor or a scale to find a length in either direction; recognise that angles are unchanged. (MA.7.GR.1.5, clarifications 1 and 3.)

**Must not** Perimeter and area under scaling are 5.02–03 ($k^2$); floor plans are 5.04; dilations on the coordinate plane are MA.8.GR.2.2; similar triangles by name are Unit 6.

**Target** I can find the scale factor between two figures, and use a scale to find a real length from a drawing or a drawing length from the real one.

**MTR** **MTR.2.1** — Notes I: the same pair of figures as a picture, two side lengths and one number k · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.5.1** — boards 1–3: one question three times

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 10 min |
| Example 1 | 8–9 | 3 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 3 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 13 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. A scale factor is the one number every length is multiplied by. Copy divided by original.
2. Less than 1 shrinks. More than 1 enlarges. The angles never change.
3. To the copy, multiply. Back to the original, divide.

## Slide by Slide

1. **Scale Drawings.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is the lesson in miniature: a scale is a ratio, and the first move is always to read it per 1. *1 retrieves 4.06 — multiply before adding; the test was yesterday, so the first band reaches to the last lesson (yesterday). 2 retrieves 4.05 — a perfect square's root; 5.02–03 uses it to find k from two areas (last week). 3 retrieves 3.08 — reading scientific notation back (last unit). 4 retrieves MA.6.AR.3.2 — a ratio as a unit rate: 12 ÷ 3. Today's scale is exactly this, read per 1 (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. −5 2. 9 3. 2,500 4. 4 km
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the figure and the line.** Two rectangles, 6 by 4 and 9 by 6.
6. **Notes.** The direction is fixed all unit: copy ÷ original.
7. **Notes.** From the original to the copy, multiply; from the copy back to the original, divide.
8. **Example 1 — Find k from two figures.** Ask which side matches which before anyone divides: 15 matches 6, 10 matches 4.
9. **Example 1 — Worked.** 15 ÷ 6 = 2.5 and 10 ÷ 4 = 2.5: the same k from both pairs, so it is a copy and k = 2.5.
10. **Your Turn — Same steps, your numbers. Boards up when done.** k = 1.5 from both pairs (12 ÷ 8 and 15 ÷ 10).
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Use a scale on a map.** Ask what the scale says per 1 inch before anyone multiplies.
13. **Example 2 — Worked.** Every inch is 25 miles, so 2.6 inches is 2.6 × 25 = 65 miles.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 63 km: 4.2 × 15.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **Board 1.** A 2 cm side becomes 6 cm in the copy. What is the scale factor? → ***k = 3*** *⅓ — divided original by copy [B1G-M 7.GR.1.5: the proportion set up backwards]; 4 — subtracted [7.GR.1.5: the additive error — Math Nation's named misconception]*
17. 
18. **Board 2.** An 8 cm side becomes 4 cm in the copy. What is the scale factor? → ***k = ½*** *2 — divided original by copy; 'k is always bigger than 1' [B1G-M 7.GR.1.5 named misconception]*
19. 
20. **Board 3.** A 12 cm side becomes 9 cm in the copy. What is the scale factor? → ***k = ¾*** *1⅓ — original ÷ copy [7.GR.1.5]; 3 — subtracted [7.GR.1.5: the additive error — Math Nation's named misconception]*
21. 
22. **Board 4.** A side of 5 cm becomes 2 cm in the copy. What is the scale factor? → ***B — 0.4*** *(A) original ÷ copy: 5 ÷ 2 [B1G-M 7.GR.1.5: the proportion set up backwards; 'k is always greater than 1']; (C) 'the same shape, so k = 1' [Math Nation 5.3: a scale factor of 1 is the same size, not the same shape]; (D) 5 − 2: the additive error — 'to*
23. 

*get from 5 to 2, take away 3' [B1G-M 7.AR.3.3: 'just add' as a conversion]  Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** The scale factor is 3.5. A side of the original is 6 cm. How long is that side in the copy? → ***21 cm*** *1.7 — divided by k going to the copy [Notes III: to the copy, multiply]*

**26–27. Board 6.** A drawing has a 40° angle. The real figure is a scale copy at $k = 2.5$. What is the angle in the real figure? → ***A — 40°*** *(B) 40 + 2.5: the additive error [7.GR.1.5: Math Nation's named misconception]; (C) 40 ÷ 2.5: the angle divided [7.GR.1.5]; (D) 40 × 2.5: the angle scaled like a length [7.GR.1.5: a scale factor multiplies lengths; the shape — its angles — is what stays the same]  Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** A triangle has sides 6, 8 and 10. Another has sides 9, 12 and 15; a third has sides 9, 12 and 14. Which one is a scale copy of the first? → ***The 9-12-15 triangle (k = 1.5); the 9-12-14 one is not a copy*** *'both' — checked one pair and stopped [Notes I: the same k from every side]; 'neither' — compared 9 to 10 instead of corresponding sides*

**30–31. Board 8.** Dev, who is 13, draws a 15-ft wall as 3 in long. How long is a 20-ft wall on his drawing? → ***4 in*** *The 13 is not needed.  100 — multiplied by 5 going to the drawing [Notes III: back to the drawing, divide]; 8 — 20 − 15 + 3, the additive error*

**32–33. Board 9.** Priya says the scale factor from a 6 cm side to its 3 cm copy is 2. Is she right? Explain, and give the scale factor. → ***No. k = 3 ÷ 6 = ½: the copy is smaller, so k is under 1.*** *'Yes, because 6 is 2 times 3' — the proportion backwards [B1G-M 7.GR.1.5]; 'No, it is 3' — subtracted*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–3 find k, going over 1, under 1, then a fraction; 4 asks k as a vote; 5 reverses the given (k and the original, asked for the copy). The pattern breaks at 6, the one measurement a scale factor leaves alone. 7 asks whether a copy is a copy at all; 8 reads a two-unit scale per 1 unit; 9 puts the direction in writing. The independent set walks the same path in six: k over 1, k under 1, k from two pairs, a side from k, a map distance, and back to the map.

## Misconceptions to Watch

**Board 1.** ⅓ — divided original by copy [B1G-M 7.GR.1.5: the proportion set up backwards]

**Board 2.** 2 — divided original by copy

**Board 3.** 1⅓ — original ÷ copy [7.GR.1.5]

**Board 4.** original ÷ copy: 5 ÷ 2 [B1G-M 7.GR.1.5: the proportion set up backwards

**Board 5.** 1.7 — divided by k going to the copy [Notes III: to the copy, multiply]

**Board 6.** 40 + 2.5: the additive error [7.GR.1.5: Math Nation's named misconception]

**Board 7.** 'both' — checked one pair and stopped [Notes I: the same k from every side]

**Board 8.** 100 — multiplied by 5 going to the drawing [Notes III: back to the drawing, divide]

**Board 9.** 'Yes, because 6 is 2 times 3' — the proportion backwards [B1G-M 7.GR.1.5]

**Materials** Whiteboards and markers. Two squares of different sizes cut from paper, for board 6's split: four right angles each. Calculators are allowed on everything and printed on nothing.

# Scale Drawings

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 5 — Multi-Step Proportional Relationships |
| Lesson | Lesson 1 — Scale Drawings | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.7.GR.1.5 |

## 2. Standards

**MA.7.GR.1.5** Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

**Must.** Find the scale factor between two figures (k = copy ÷ original, in either direction of size) and use a scale factor or a scale to find a length in either direction; recognise that angles are unchanged. (MA.7.GR.1.5, clarifications 1 and 3.)

**Must not.** Perimeter and area under scaling are 5.02–03 ($k^2$); floor plans are 5.04; dilations on the coordinate plane are MA.8.GR.2.2; similar triangles by name are Unit 6.

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes I: the same pair of figures as a picture, two side lengths and one number k |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote |
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts. | boards 1–3: one question three times — k over 1, under 1, a fraction |

## 4. Learning Target and Essential Question

**Learning target.** I can find the scale factor between two figures, and use a scale to find a real length from a drawing or a drawing length from the real one.

**Essential question.** If a copy's sides are 6 and 9 where the original's were 4 and 6, what one number was every side multiplied by?

**Building on.** Grade 6 (MA.6.AR.3.2) — a ratio as a unit rate; Unit 4 — multiplying and dividing decimals and fractions.

**Working toward.** 5.02–03 — perimeter by k, area by $k^2$; 5.04 — floor plans and maps; Unit 6 — similar triangles are scale copies.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front; say the off-slide sentence out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 13 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; due by the Unit 5 test. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Two rectangles are 4 by 6 and 8 by 10. Why is the second not a scale copy of the first? | DOK 3 |
| A copy is smaller than its original. What do you know about k before you compute it? | DOK 2 |
| Every side of a triangle is tripled. Why can none of its angles change? | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | A 2 cm side becomes 6 cm in the copy. What is the scale factor? | ***k = 3*** | ⅓ — divided original by copy [B1G-M 7.GR.1.5: the proportion set up backwards]; 4 — subtracted [7.GR.1.5: the additive error — Math Nation's named misconception] |
| 2 | An 8 cm side becomes 4 cm in the copy. What is the scale factor? | ***k = ½*** | 2 — divided original by copy; 'k is always bigger than 1' [B1G-M 7.GR.1.5 named misconception] |
| 3 | A 12 cm side becomes 9 cm in the copy. What is the scale factor? | ***k = ¾*** | 1⅓ — original ÷ copy [7.GR.1.5]; 3 — subtracted [7.GR.1.5: the additive error — Math Nation's named misconception] |
| 4 | A side of 5 cm becomes 2 cm in the copy. What is the scale factor? | ***B — 0.4*** | (A) original ÷ copy: 5 ÷ 2 [B1G-M 7.GR.1.5: the proportion set up backwards; 'k is always greater than 1']; (C) 'the same shape, so k = 1' [Math Nation 5.3: a scale factor of 1 is the same size, not the same shape]; (D) 5 − 2: the additive error — 'to get from 5 to 2, take away 3' [B1G-M 7.AR.3.3: 'just add' as a conversion] |
| 5 | The scale factor is 3.5. A side of the original is 6 cm. How long is that side in the copy? | ***21 cm*** | 1.7 — divided by k going to the copy [Notes III: to the copy, multiply] |
| 6 | A drawing has a 40° angle. The real figure is a scale copy at $k = 2.5$. What is the angle in the real figure? | ***A — 40°*** | (B) 40 + 2.5: the additive error [7.GR.1.5: Math Nation's named misconception]; (C) 40 ÷ 2.5: the angle divided [7.GR.1.5]; (D) 40 × 2.5: the angle scaled like a length [7.GR.1.5: a scale factor multiplies lengths; the shape — its angles — is what stays the same] |
| 7 | A triangle has sides 6, 8 and 10. Another has sides 9, 12 and 15; a third has sides 9, 12 and 14. Which one is a scale copy of the first? | ***The 9-12-15 triangle (k = 1.5); the 9-12-14 one is not a copy*** | 'both' — checked one pair and stopped [Notes I: the same k from every side]; 'neither' — compared 9 to 10 instead of corresponding sides |
| 8 | Dev, who is 13, draws a 15-ft wall as 3 in long. How long is a 20-ft wall on his drawing? | ***4 in*** | 100 — multiplied by 5 going to the drawing [Notes III: back to the drawing, divide]; 8 — 20 − 15 + 3, the additive error |
| 9 | Priya says the scale factor from a 6 cm side to its 3 cm copy is 2. Is she right? Explain, and give the scale factor. | ***No. k = 3 ÷ 6 = ½: the copy is smaller, so k is under 1.*** | 'Yes, because 6 is 2 times 3' — the proportion backwards [B1G-M 7.GR.1.5]; 'No, it is 3' — subtracted |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** The direction is written on the board all period as one line — copy ÷ original — and every k is found by writing the copy's number over the original's before dividing. The two rectangles of Notes I stay up beside it.

**ELL.** **Scale factor** is one number with no unit; **scale** has units on it (1 in : 20 mi). Say "copy divided by original" with every k, and point to which figure is which.

**Enrichment.** Board 7 finishes early for some: ask them for a fourth triangle with sides that are a scale copy of 6-8-10 at k = 2.5, and whether its angles changed.

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the student has to say which way the division goes (copy ÷ original) and give k = ½. A board that says 'yes, 6 is 2 times 3' has the direction backwards — the unit's first misconception, and tomorrow's warm-up Q1 asks it again.

**Vocabulary.** **scale drawing**; **scale factor**; **corresponding sides**; **scale**

**Materials.** Whiteboards and markers. Two squares of different sizes cut from paper, for board 6's split: four right angles each. Calculators are allowed on everything and printed on nothing.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 5 test: Scale drawings: find the scale factor, missing side, or missing angle — WEA.

# Scale Factor: Perimeter and Area

*Teacher's Edition*

## The Period

**MA.7.GR.1.5** Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

**Must** Find the perimeter and area of a scale copy from the original and k (perimeter × k, area × $k^2$), and find k from two areas when the ratio is a perfect square. (MA.7.GR.1.5, clarification 2.)

**Must not** Area ratios that are not perfect squares (an approximate root); the area of a figure the class has no formula for; area and perimeter in different units (5.04 and 5.08–09).

**Target** I can find the perimeter and the area of a scale copy from the original and k, and find k from two areas.

**MTR** **MTR.2.1** — Notes II: the same copy as a grid picture, as 15 → 60, and as $k^2 = 4$ · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote on k against $k^2$ · **MTR.5.1** — boards 1–2: the same k on the perimeter, then the area, and the room is asked why the second answer is bigger

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 10 min |
| Example 1 | 8–9 | 4 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 11 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. The perimeter is lengths, so it is multiplied by k.
2. Area is two lengths multiplied, so it is multiplied by k twice: k squared.
3. From two areas, k is the square root of the ratio.

## Slide by Slide

1. **Scale Factor: Perimeter and Area.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is the figure Notes I and II scale: 3 by 5, P = 16, A = 15. *1 retrieves 5.01 — copy ÷ original; k under 1 (yesterday's closure promised it) (yesterday). 2 retrieves 4.05 — a perfect square's root; Notes III finds k this way (last week). 3 retrieves 3.01 — powers; $k^2$ is a power (last unit). 4 retrieves MA.6.GR.2.1 — perimeter is the sum of the sides, area is length × width; today both are scaled (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. k = 0.4 2. 12 3. 9 and 27 4. P = 16 cm, A = 15 $cm^2$
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the figure and the two lines.** The 3 by 5 from the warm-up at k = 2: 6 by 10.
6. **Notes.** Area is length × width, and both were multiplied by k, so the area is multiplied by k twice: 15 → 60, four times as much, not twice.
7. **Notes.** Run it backwards.
8. **Example 1 — Perimeter and area from k.** Ask which rule each part uses before anyone multiplies: perimeter × k, area × $k^2$.
9. **Example 1 — Worked.** P = 12 → 36 (× 3); A = 8 → 72 (× 9).
10. **Your Turn — Same steps, your numbers. Boards up when done.** P 18 → 72, A 18 → 288.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — k from two areas, then a side.** Ask what 180 ÷ 20 is, and whether that number is k.
13. **Example 2 — Worked.** 240 ÷ 15 = 16 is $k^2$, so k = 4, and the side is 12.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 300 ÷ 12 = 25, k = 5, the side is 10.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **Board 1.** A figure with perimeter 14 cm is scaled by $k=2$. What is the copy's perimeter? → ***28 cm*** *56 — perimeter × $k^2$ [Notes I: a perimeter is lengths, × k]; 16 — added k [7.GR.1.5: the additive error — Math Nation's named misconception]*

    **17. Board 1.**
17. **Board 2.** A figure with area $12\,cm^2$ is scaled by $k=2$. What is the copy's area? → ***48 cm²*** *24 — area × k [B1G-M 7.GR.1.5 named misconception: scaling area like length]*

    **18–19. Board 2.**
18. **Board 3.** A figure with area $48\,cm^2$ is scaled by $k=\frac{1}{2}$. What is the copy's area? → ***12 cm²*** *24 — area × k [7.GR.1.5: area × $k^2$]; 96 — divided by ½ as if k were 2 [5.01: k under 1 shrinks]*

    **19–20. Board 3.**
19. **Board 4.** A figure with area $10\,cm^2$ is scaled by $k=3$. What is the copy's area? → ***A — 90 cm²*** *(B) area × 2k: 'two lengths, so twice k' [Notes II: k × k, not k + k]; (C) area × k: the guide's named misconception [B1G-M 7.GR.1.5: scaling area like length]; (D) 10 + 3: the additive error [7.GR.1.5: Math Nation's named misconception]*

    **20–21. Board 4.**

*Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** A figure with area $8\,\text{cm}^2$ is scaled by $k = 1.5$. What is the copy's area? → ***18 cm²*** *12 — area × k [7.GR.1.5]; 24 — area × 2k [Notes II: k × k]*

**26–27. Board 6.** A figure's perimeter is 20 cm; its copy's perimeter is 60 cm. What is the area ratio, copy to original? → ***D — 9*** *(A) 2k: 'area has two dimensions, so double it' [Notes II]; (B) the perimeter ratio handed in as the area ratio [Notes I and II: perimeter gives k, area is k²]; (C) 60 − 20: the additive error [7.GR.1.5: Math Nation's named misconception] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** A triangle with sides 6, 8 and 10 cm is scaled by $k = \frac{1}{2}$. Its area is $24\,\text{cm}^2$. Find the copy's perimeter and area. → ***P = 12 cm, A = 6 cm²*** *P 12, A 12 — the area halved like a length [7.GR.1.5]; P 6, A 6 — both quartered*

**30–31. Board 8.** A 4 in by 6 in photo, taken 3 years ago, is enlarged at $k = 2.5$. What is the enlargement's area? → ***150 in²*** *The 3 years is not needed. 60 — area × k [7.GR.1.5]; 120 — 24 × 5, k doubled [Notes II]*

**32–33. Board 9.** Jo says that doubling every side of a rectangle doubles its area. Explain why not, with a counterexample. → ***No: 2 by 7 doubled is 4 by 14, area 14 → 56 — × 4, as both lengths doubled.*** *'Yes, because every side doubles' — scaling area like length [B1G-M 7.GR.1.5]; 'No, it is k²' with no numbers — the rule without the reason*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2 keep k = 2 and move from perimeter to area; 3 takes k under 1; 4 asks the area as a vote; 5 makes k a decimal. The pattern breaks at 6, where k comes from perimeters and the area ratio is asked. 7 is a triangle, 8 the extra number, 9 the reason in writing. The independent set walks the same path: perimeter × k, area × k², k under 1, both on one rectangle, k from areas, the area ratio from perimeters.

## Misconceptions to Watch

**Board 1.** 56 — perimeter × k² [Notes I: a perimeter is lengths, × k]

**Board 2.** 24 — area × k [B1G-M 7.GR.1.5 named misconception: scaling area like length]

**Board 3.** 24 — area × k [7.GR.1.5: area × k²]

**Board 4.** area × 2k: 'two lengths, so twice k' [Notes II: k × k, not k + k]

**Board 5.** 12 — area × k [7.GR.1.5]

**Board 6.** 2k: 'area has two dimensions, so double it' [Notes II]

**Board 7.** P 12, A 12 — the area halved like a length [7.GR.1.5]

**Board 8.** 60 — area × k [7.GR.1.5]

**Board 9.** 'Yes, because every side doubles' — scaling area like length [B1G-M 7.GR.1.5]

**Materials** Whiteboards and markers. A sheet of grid paper under the document camera for Notes II's four-copies picture, drawn live if the slide's grid is not enough. Calculators are allowed on everything and printed on nothing.

# Scale Factor: Perimeter and Area

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 5 — Multi-Step Proportional Relationships |
| Lesson | Lessons 2–3 — Scale Factor: Perimeter and Area | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.7.GR.1.5 |

## 2. Standards

**MA.7.GR.1.5** Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

**Must.** Find the perimeter and area of a scale copy from the original and k (perimeter × k, area × $k^2$), and find k from two areas when the ratio is a perfect square. (MA.7.GR.1.5, clarification 2.)

**Must not.** Area ratios that are not perfect squares (an approximate root); the area of a figure the class has no formula for; area and perimeter in different units (5.04 and 5.08–09).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes II: the same copy as a grid picture, as 15 → 60, and as $k^2 = 4$ |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote on k against $k^2$ |
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts. | boards 1–2: the same k on the perimeter, then the area, and the room is asked why the second answer is bigger |

## 4. Learning Target and Essential Question

**Learning target.** I can find the perimeter and the area of a scale copy from the original and k, and find k from two areas.

**Essential question.** Every side of a rectangle is doubled. Why is its area four times as big, not twice?

**Building on.** 5.01 — k = copy ÷ original; 4.05 — square roots of perfect squares; Grade 6 — perimeter and area of a rectangle.

**Working toward.** 5.04 — floor plans, where area has to be found in real units; 5.08–09 — area converts by the square of the length ratio.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front; say the off-slide sentence out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; due by the Unit 5 test. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| A rectangle's sides are tripled. Its perimeter is multiplied by 3 and its area by 9. Why are the two numbers different? | DOK 3 |
| Two squares have areas 25 and 100. What is k, and how do you know without finding a side? | DOK 2 |
| Could a copy have the same perimeter as its original and a different area? Explain. | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | A figure with perimeter 14 cm is scaled by $k = 2$. What is the copy's perimeter? | ***28 cm*** | 56 — perimeter × $k^2$ [Notes I: a perimeter is lengths, × k]; 16 — added k [7.GR.1.5: the additive error — Math Nation's named misconception] |
| 2 | A figure with area $12\,\text{cm}^2$ is scaled by $k = 2$. What is the copy's area? | ***48 cm²*** | 24 — area × k [B1G-M 7.GR.1.5 named misconception: scaling area like length] |
| 3 | A figure with area $48\,\text{cm}^2$ is scaled by $k = \frac{1}{2}$. What is the copy's area? | ***12 cm²*** | 24 — area × k [7.GR.1.5: area × $k^2$]; 96 — divided by ½ as if k were 2 [5.01: k under 1 shrinks] |
| 4 | A figure with area $10\,\text{cm}^2$ is scaled by $k = 3$. What is the copy's area? | ***A — 90 cm²*** | (B) area × 2k: 'two lengths, so twice k' [Notes II: k × k, not k + k]; (C) area × k: the guide's named misconception [B1G-M 7.GR.1.5: scaling area like length]; (D) 10 + 3: the additive error [7.GR.1.5: Math Nation's named misconception] |
| 5 | A figure with area $8\,\text{cm}^2$ is scaled by $k = 1.5$. What is the copy's area? | ***18 cm²*** | 12 — area × k [7.GR.1.5]; 24 — area × 2k [Notes II: k × k] |
| 6 | A figure's perimeter is 20 cm; its copy's perimeter is 60 cm. What is the area ratio, copy to original? | ***D — 9*** | (A) 2k: 'area has two dimensions, so double it' [Notes II]; (B) the perimeter ratio handed in as the area ratio [Notes I and II: perimeter gives k, area is $k^2$]; (C) 60 − 20: the additive error [7.GR.1.5: Math Nation's named misconception] |
| 7 | A triangle with sides 6, 8 and 10 cm is scaled by $k = \frac{1}{2}$. Its area is $24\,\text{cm}^2$. Find the copy's perimeter and area. | ***P = 12 cm, A = 6 cm²*** | P 12, A 12 — the area halved like a length [7.GR.1.5]; P 6, A 6 — both quartered |
| 8 | A 4 in by 6 in photo, taken 3 years ago, is enlarged at $k = 2.5$. What is the enlargement's area? | ***150 in²*** | 60 — area × k [7.GR.1.5]; 120 — 24 × 5, k doubled [Notes II] |
| 9 | Jo says that doubling every side of a rectangle doubles its area. Explain why not, with a counterexample. | ***No: 2 by 7 doubled is 4 by 14, area 14 → 56 — × 4, as both lengths doubled.*** | 'Yes, because every side doubles' — scaling area like length [B1G-M 7.GR.1.5]; 'No, it is $k^{2}$' with no numbers — the rule without the reason |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** Two lines stay on the board all period: perimeter × k, area × k × k. Every board is answered by pointing at one of them before multiplying. The grid picture of Notes II stays up beside them.

**ELL.** **Perimeter** is the distance around; **area** is the squares inside. Say "around" and "inside" with each board, and the word **squared** means times itself.

**Enrichment.** Board 7 finishes early for some: ask them what k does to a triangle's angles (nothing) and to its height (× k), and why the area rule still holds when there is no length × width.

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: a sentence that area is two lengths multiplied, so it is multiplied by k twice, with a counterexample's numbers (a 2 by 7 at k = 2: 14 → 56, not 28). A board that only says 'because $k^2$' has named the rule, not the reason.

**Vocabulary.** **k squared**; **perimeter ratio**; **area ratio**

**Materials.** Whiteboards and markers. A sheet of grid paper under the document camera for Notes II's four-copies picture, drawn live if the slide's grid is not enough. Calculators are allowed on everything and printed on nothing.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 5 test: Perimeter and area: changes in scale — ZC6; Scale drawings: find the scale factor or missing side — XHM.

# Real-World Scale Drawings

*Teacher's Edition*

## The Period

**MA.7.GR.1.5** Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

**Must** Read a scale such as ¼ in : 1 ft or 1 : 50 as a rate, find real lengths and drawing lengths from it in both directions, and find a real area from the real dimensions. (MA.7.GR.1.5, clarifications 1 and 3; B1G-M's wall example.)

**Must not** Converting the real area into another system (5.08–09); drawing a plan to scale with a ruler (not a FAST shape — the audit's ruler items are out); a scale factor between two drawings of one object (5.01).

**Target** I can use a scale such as ¼ in : 1 ft to find real lengths and areas from a drawing, and drawing lengths from real ones.

**MTR** **MTR.1.1** — Example 2 and its Your Turn: the scale is not printed, and the room has to make it from one pair before the question can start · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.7.1** — every board is a plan, a map or a model: the scale is a real object's, and the answer is checked against what a room or a car could be

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 10 min |
| Example 1 | 8–9 | 4 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 11 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. A scale with two units is a rate: write it per 1 unit of drawing first.
2. Drawing × rate is real; real ÷ rate is drawing.
3. Area comes from the real dimensions — the rate is used twice, never once.

## Slide by Slide

1. **Real-World Scale Drawings.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is the move Notes I makes on a scale: per 1 unit first. *1 retrieves 5.02–03 — area × $k^2$ (yesterday). 2 retrieves 5.01 — copy ÷ original; k under 1 (earlier this unit). 3 retrieves 4.03 — decimal products; every board today has one (last unit). 4 retrieves MA.6.AR.3.2 — a ratio written as a unit rate; today a scale is read the same way (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 36 $cm^2$ 2. k = 0.6 3. 3.6 4. 45 miles per hour
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the scale and its rewrite.** ¼ in : 1 ft says a quarter inch of paper stands for a foot of room.
6. **Notes.** A room 2 in by 2.5 in at 1 in : 4 ft is 8 ft by 10 ft, so 80 $ft^2$.
7. **Notes.** A 14-ft wall at 1 in : 4 ft: 14 ÷ 4 = 3.5 in on the paper.
8. **Example 1 — A room's real dimensions and area.** Ask for the scale per 1 inch before anyone multiplies.
9. **Example 1 — Worked.** 1 in : 4 ft; 10 ft by 12 ft; 120 $ft^2$.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 6 by 16; 96 $ft^2$.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — The scale is not printed: make it from one pair.** Ask what the scale is before anyone touches the 6 — nobody multiplies until it is written per 1 inch.
13. **Example 2 — Worked.** The pair gives the scale: 18 ÷ 4 = 4.5 ft per inch.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 6.5 ft per inch; 32.5 ft.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

**16–17. Board 1.** A plan is drawn at $\frac{1}{2}\text{ in} : 1\text{ ft}$. A wall is 4 in long on it. How long is the real wall? → ***8 ft*** *2 — divided (4 × ½) [Notes I: per 1 inch first, then × 2]; 4 — read the scale as 1 : 1*

**18–19. Board 2.** A plan is drawn at $1\text{ in} : 3\text{ ft}$. A room is 2.5 in long on it. How long is the real room? → ***7.5 ft*** *5.5 — added 3 [7.GR.1.5: the additive error — Math Nation's named misconception]; 0.83 — divided [Notes I: drawing to real is × rate]*

**20–21. Board 3.** A plan is drawn at $1\text{ in} : 3\text{ ft}$. A real wall is 21 ft long. How long is it on the plan? → ***7 in*** *63 — multiplied [Notes III: real to drawing is a division]; 18 — subtracted 3*

**22–23. Board 4.** A model car is built at $1 : 50$ and is 8 cm long. How long is the real car? → ***A — 4 m*** *(B) 8 ÷ 50: divided [Notes I: drawing to real is × rate]; (C) 400 with the wrong unit: 400 cm, not 400 m [Notes I: the scale's units are the drawing's]; (D) 8 + 50: the additive error [B1G-M 7.GR.1.5] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** A plan is drawn at $\frac{1}{4}\,\text{in} : 1\,\text{ft}$. A real wall is 15 ft long. How long is it on the plan? → ***3.75 in*** *60 — multiplied by 4 [Notes III: real to drawing is ÷]; 7.5 — divided by 2*

**26–27. Board 6.** A plan is drawn at $1\,\text{in} : 4\,\text{ft}$. A room is 2 in by 3 in on it. What is the room's real area? → ***B — 96 ft²*** *(A) 6 in² × 4: the drawing's area scaled by the length rate [B1G-M 7.GR.1.5: scaling area like length; Notes II]; (C) 2(8 + 12): the real perimeter, not the area [5.02–03 Notes I]; (D) 8 + 12: the real dimensions added [7.GR.1.5: the additive error — Math Nation's named misconception] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** A plane is 36 ft long. One model of it is built at $1:48$, another at $1:72$. Which model is longer, and by how much? → ***The 1 : 48 model, by 3 in (9 in against 6 in)*** *the 1 : 72 model — 'the bigger scale is the bigger model' [Notes III: real ÷ rate — a bigger rate divides more]; 0.75 ft and 0.5 ft with no comparison — stopped early*

**30–31. Board 8.** Mia, who is 13, enlarges her 9 in by 12 in poster into a mural. The mural's scale is $1\,\text{in} : 2\,\text{ft}$. What are the mural's dimensions? → ***18 ft by 24 ft*** *The 13 (her age) is not needed. 4.5 ft by 6 ft — divided [Notes I: drawing to real is × rate]; 18 in by 24 in — kept the drawing's unit*

**32–33. Board 9.** On Andrew's drawing a wall is 3.5 in wide and 4 in tall; the real wall is 26.25 ft wide. Find the real height and the real area, and say which step uses the scale. → ***30 ft high; 787.5 ft². The scale (7.5 ft per in) is the step from 4 in to 30 ft.*** *105 ft² — 3.5 × 4 = 14 in², then × 7.5 [B1G-M 7.GR.1.5: scaling area like length]; 'the area step uses the scale' — the scale is between a drawing and a real length, not two real ones*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2 take a drawing length to a real one, the scale changing once; 3 reverses the direction; 4 is a one-unit model scale as a vote; 5 is the quarter-inch scale back to the paper. The pattern breaks at 6, where area is asked and the room has to go to real dimensions before multiplying. 7 compares two models of one object, 8 is the extra number, 9 the B1G-M's wall in writing. The independent set walks the same path: two scales, both directions, a real area, a model in meters, a scale made from one pair.

## Misconceptions to Watch

**Board 1.** 2 — divided (4 × ½) [Notes I: per 1 inch first, then × 2]

**Board 2.** 5.5 — added 3 [7.GR.1.5: the additive error — Math Nation's named misconception]

**Board 3.** 63 — multiplied [Notes III: real to drawing is a division]

**Board 4.** 8 ÷ 50: divided [Notes I: drawing to real is × rate]

**Board 5.** 60 — multiplied by 4 [Notes III: real to drawing is ÷]

**Board 6.** 6 in² × 4: the drawing's area scaled by the length rate [B1G-M 7.GR.1.5: scaling area like length

**Board 7.** the 1 : 72 model — 'the bigger scale is the bigger model' [Notes III: real ÷ rate — a bigger rate divides more]

**Board 8.** 4.5 ft by 6 ft — divided [Notes I: drawing to real is × rate]

**Board 9.** 105 ft² — 3.5 × 4 = 14 in², then × 7.5 [B1G-M 7.GR.1.5: scaling area like length]

**Materials** Whiteboards and markers. A printed floor plan under the document camera for Notes II if the slide's rectangle is not enough. Calculators are allowed on everything and printed on nothing.

# Real-World Scale Drawings

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 5 — Multi-Step Proportional Relationships |
| Lesson | Lesson 4 — Real-World Scale Drawings | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.7.GR.1.5 |

## 2. Standards

**MA.7.GR.1.5** Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

**Must.** Read a scale such as ¼ in : 1 ft or 1 : 50 as a rate, find real lengths and drawing lengths from it in both directions, and find a real area from the real dimensions. (MA.7.GR.1.5, clarifications 1 and 3; B1G-M's wall example.)

**Must not.** Converting the real area into another system (5.08–09); drawing a plan to scale with a ruler (not a FAST shape — the audit's ruler items are out); a scale factor between two drawings of one object (5.01).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.1.1 | Actively participate in effortful learning both individually and collectively. | Example 2 and its Your Turn: the scale is not printed, and the room has to make it from one pair before the question can start |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote |
| MA.K12.MTR.7.1 | Apply mathematics to real-world contexts. | every board is a plan, a map or a model: the scale is a real object's, and the answer is checked against what a room or a car could be |

## 4. Learning Target and Essential Question

**Learning target.** I can use a scale such as ¼ in : 1 ft to find real lengths and areas from a drawing, and drawing lengths from real ones.

**Essential question.** A room is 2 in by 3 in on a plan at 1 in : 4 ft. Why is its real area not 6 × 4 square feet?

**Building on.** 5.01 — a map scale used both ways; 5.02–03 — area is two lengths, so the rate is used twice; Grade 6 — a ratio written as a unit rate.

**Working toward.** 5.06–07 — a conversion ratio is a rate too, and it is multiplied the same way; 5.10–11 — a scale, then a speed: two rates in one problem.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front; say the off-slide sentence out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; due by the Unit 5 test. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| A plan's scale is ¼ in : 1 ft. Why does writing it as 1 in : 4 ft make every question easier? | DOK 2 |
| A room is 2 in by 3 in on that plan. Why is its real area 96 $ft^2$ and not 24? | DOK 3 |
| Two models of one plane are built at 1 : 48 and 1 : 72. Which is longer, and why does the bigger number make the smaller model? | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | A plan is drawn at $\frac{1}{2}\text{ in} : 1\text{ ft}$. A wall is 4 in long on it. How long is the real wall? | ***8 ft*** | 2 — divided (4 × ½) [Notes I: per 1 inch first, then × 2]; 4 — read the scale as 1 : 1 |
| 2 | A plan is drawn at $1\text{ in} : 3\text{ ft}$. A room is 2.5 in long on it. How long is the real room? | ***7.5 ft*** | 5.5 — added 3 [7.GR.1.5: the additive error — Math Nation's named misconception]; 0.83 — divided [Notes I: drawing to real is × rate] |
| 3 | A plan is drawn at $1\text{ in} : 3\text{ ft}$. A real wall is 21 ft long. How long is it on the plan? | ***7 in*** | 63 — multiplied [Notes III: real to drawing is a division]; 18 — subtracted 3 |
| 4 | A model car is built at $1 : 50$ and is 8 cm long. How long is the real car? | ***A — 4 m*** | (B) 8 ÷ 50: divided [Notes I: drawing to real is × rate]; (C) 400 with the wrong unit: 400 cm, not 400 m [Notes I: the scale's units are the drawing's]; (D) 8 + 50: the additive error [B1G-M 7.GR.1.5] |
| 5 | A plan is drawn at $\frac{1}{4}\text{ in} : 1\text{ ft}$. A real wall is 15 ft long. How long is it on the plan? | ***3.75 in*** | 60 — multiplied by 4 [Notes III: real to drawing is ÷]; 7.5 — divided by 2 |
| 6 | A plan is drawn at $1\text{ in} : 4\text{ ft}$. A room is 2 in by 3 in on it. What is the room's real area? | ***B — 96 ft²*** | (A) 6 in² × 4: the drawing's area scaled by the length rate [B1G-M 7.GR.1.5: scaling area like length; Notes II]; (C) 2(8 + 12): the real perimeter, not the area [5.02–03 Notes I]; (D) 8 + 12: the real dimensions added [7.GR.1.5: the additive error — Math Nation's named misconception] |
| 7 | A plane is 36 ft long. One model of it is built at $1 : 48$, another at $1 : 72$. Which model is longer, and by how much? | ***The 1 : 48 model, by 3 in (9 in against 6 in)*** | the 1 : 72 model — 'the bigger scale is the bigger model' [Notes III: real ÷ rate — a bigger rate divides more]; 0.75 ft and 0.5 ft with no comparison — stopped early |
| 8 | Mia, who is 13, enlarges her 9 in by 12 in poster into a mural. The mural's scale is $1\text{ in} : 2\text{ ft}$. What are the mural's dimensions? | ***18 ft by 24 ft*** | 4.5 ft by 6 ft — divided [Notes I: drawing to real is × rate]; 18 in by 24 in — kept the drawing's unit |
| 9 | On Andrew's drawing a wall is 3.5 in wide and 4 in tall; the real wall is 26.25 ft wide. Find the real height and the real area, and say which step uses the scale. | ***30 ft high; 787.5 ft². The scale (7.5 ft per in) is the step from 4 in to 30 ft.*** | 105 ft² — 3.5 × 4 = 14 in², then × 7.5 [B1G-M 7.GR.1.5: scaling area like length]; 'the area step uses the scale' — the scale is between a drawing and a real length, not two real ones |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** Every scale is rewritten per 1 drawing unit on the board before a question is read, and stays there. Two lines under it all period: drawing × rate = real; real ÷ rate = drawing.

**ELL.** **Scale** here is a ratio, not the kitchen scale; **plan** means a drawing from above. Say "on the paper" for drawing lengths and "in the room" for real ones with every board.

**Enrichment.** Board 7 finishes early for some: ask them to write the two models' lengths in centimeters with 1 in = 2.54 cm — Monday's lesson, a day early.

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the scale made from the pair (7.5 ft per inch, as Example 2 did), the height from it (30 ft), the area from the two real dimensions (787.5 $ft^2$), and a sentence naming which step uses the scale. A board that scales $3.5 \times 4 = 14$ $in^2$ by 7.5 has not gone to real units first.

**Vocabulary.** **floor plan**; **scale (two units)**; **unit rate of a scale**; **scale with one unit**

**Materials.** Whiteboards and markers. A printed floor plan under the document camera for Notes II if the slide's rectangle is not enough. Calculators are allowed on everything and printed on nothing.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 5 test: Scale drawings: word problems — 84H; Scale drawings: scale factor word problems — KCM.

# Converting Length Across Systems; Currency

*Teacher's Edition*

## The Period

**MA.7.AR.3.3** Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

**Must** Convert a length between customary and metric with a printed reference-sheet ratio, either direction, in a chain; convert money with a given rate, both ways. (MA.7.AR.3.3.)

**Must not** A ratio not on the FAST reference sheet; a real exchange rate the student must know; mass, volume or area (5.08–09).

**Target** I can convert a length between the customary and metric systems with a ratio from the reference sheet, and convert money with a given rate.

**MTR** **MTR.6.1** — Notes II and board 6: the estimate before the arithmetic · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote on which ratio · **MTR.3.1** — boards 5–6: two conversions in a row, each one a form of 1, with the units cancelling on the paper

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 10 min |
| Example 1 | 8–9 | 4 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 11 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. A conversion ratio is a form of 1: multiply by it and only the unit changes.
2. Pick the ratio whose 1 is the unit you have. Estimate first — which unit is bigger?
3. Money is the same move with a rate someone sets, printed on the problem.

## Slide by Slide

1. **Converting Length Across Systems; Currency.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is the move: a ratio, and a division because feet are bigger. *1 retrieves 5.04 — real ÷ rate (last class). 2 retrieves 5.01 — copy ÷ original; k under 1 (last week). 3 retrieves 4.02 — decimal quotients (last unit). 4 retrieves MA.6.AR.3.5 — converting inside one system; today the ratio crosses systems (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 2.5 in 2. k = ½ 3. 5 4. 2.5 ft
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the ratio and the line.** 1 in = 2.54 cm, so 2.54 cm over 1 in is 1 — the top and bottom are the same length in two units.
6. **Notes.** The sheet gives both directions: 1 in = 2.54 cm and 1 cm = 0.39 in.
7. **Notes.** An exchange rate is a conversion ratio with a date on it: 0.92 euros per dollar today, something else tomorrow, so it is printed on the problem and never on the sheet.
8. **Example 1 — Inches to centimeters.** Ask which unit should give the bigger number before anyone multiplies.
9. **Example 1 — Worked.** 38.1 cm.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 63.5 cm.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Dollars to euros, and back.** Ask what the second question's operation is before anyone writes it.
13. **Example 2 — Worked.** 73.60 euros; $50.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 162 CAD; $60.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** $1\,\text{in} = 2.54\,\text{cm}$ . A ruler is 20 in long. How many centimeters is that? → ***50.8 cm*** *7.87 — divided by 2.54 [Notes II: centimeters are smaller, so more of them]; 22.54 — added the ratio*
17. **18–19. Board 2.** $1\,\text{cm} = 0.39\,\text{in}$ . A phone is 20 cm long. How many inches is that? → ***7.8 in*** *50.8 — multiplied by 2.54 [Notes II: pick the ratio whose 1 is the unit you have]; 0.78 — place value*
18. **20–21. Board 3.** $1\,\text{mi} = 1.61\,\text{km}$ . A race is 5 mi long. How many kilometers is that? → ***8.05 km*** *3.1 — multiplied by 0.62, the other ratio [Notes II]; 6.61 — added 1.61*
19. **22–23. Board 4.** $1\,\text{km} = 0.62\,\text{mi}$ . A trail is 5 km long. How many miles is that? → ***D — 3.1 mi*** *(A) 5 + 0.62: the additive error [7.AR.3.3: Math Nation's named misconception]; (B) place value: 5 × 0.62 is 3.1, not 0.31 [4.03]; (C) × 1.61, the ratio for the other direction [B1G-M 7.AR.3.3: the inverted ratio; Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*
20. **24–25. Board 5.** $1\,\text{ft} = 12\,\text{in}$ and $1\,\text{in} = 2.54\,\text{cm}$ . A door is 6 ft tall. How many centimeters is that? → ***182.88 cm*** *15.24 — 6 × 2.54, the feet treated as inches*

*[Notes I: the unit you have must match the ratio's 1]; 183 — rounded early (accept with the work shown)*

**26–27. Board 6.** $1\text{ yd} = 3\text{ ft}$ , $1\text{ ft} = 12\text{ in}$ , $1\text{ in} = 2.54\text{ cm}$ . A table is 3 yd long. About how many centimeters is that? Estimate, then compute. → ***About 270; exactly 274.32 cm*** *7.62 — 3 × 2.54, the yards treated as inches [Notes II: estimate first — a 3-yd table is not 8 cm]; 91.44 — stopped at 36 in × 2.54 (one yard)*

**28–29. Board 7.** Rate on Nov 2: $\$1 = 1.32\text{ CAD}$ ; $1\text{ CAD} = 16.2\text{ pesos}$ . Theo changes $200 to CAD, then to pesos. How many pesos does he get? → ***4,276.80 pesos*** *3,240 — 200 × 16.2, the first hop skipped [Notes III: the rate's 1 must be the currency you have]; 2,454.55 — divided by 1.32 on the first hop*

**30–31. Board 8.** $1\text{ km} = 0.62\text{ mi}$ . Theo, who weighs 110 lb, rides a 12 km trail. How long is the trail in miles? → ***7.44 mi*** *The 110 lb is not needed. 19.32 — × 1.61 [B1G-M 7.AR.3.3: the inverted ratio]; 49.94 kg — converted the weight instead*

**32–33. Board 9.** $1\text{ in} = 2.54\text{ cm}$ and $1\text{ cm} = 0.39\text{ in}$ . Jo converts 30 cm to inches by multiplying by 2.54 and gets 76.2 in. Name the error and fix it. → ***She used the inches ratio on centimeters. Inches are bigger, so fewer: 11.7 in.*** *'She should have used 0.39' with no reason — the fix without the error named; 76.2 defended as 'more inches' — the estimate backwards [B1G-M 7.AR.3.3]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2 take one ratio both ways; 3–4 the next ratio, the second as a vote; 5 chains inside customary then across. The pattern breaks at 6, where three conversions run in a row and the estimate is the only check. 7 is a currency chain, 8 the extra number, 9 the error in writing. The independent set walks the same path: the four length ratios, a chain with an estimate, a currency round trip.

## Misconceptions to Watch

**Board 1.** 7.87 — divided by 2.54 [Notes II: centimeters are smaller, so more of them]

**Board 2.** 50.8 — multiplied by 2.54 [Notes II: pick the ratio whose 1 is the unit you have]

**Board 3.** 3.1 — multiplied by 0.62, the other ratio [Notes II]

**Board 4.** 5 + 0.62: the additive error [7.AR.3.3: Math Nation's named misconception]

**Board 5.** 15.24 — 6 × 2.54, the feet treated as inches [Notes I: the unit you have must match the ratio's 1]

**Board 6.** 7.62 — 3 × 2.54, the yards treated as inches [Notes II: estimate first — a 3-yd table is not 8 cm]

**Board 7.** 3,240 — 200 × 16.2, the first hop skipped [Notes III: the rate's 1 must be the currency you have]

**Board 8.** 19.32 — × 1.61 [B1G-M 7.AR.3.3: the inverted ratio]

**Board 9.** 'She should have used 0.39' with no reason — the fix without the error named

**Materials** Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (they are printed on every item as well). Calculators are allowed on everything and printed on nothing.

# Converting Length Across Systems; Currency

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 5 — Multi-Step Proportional Relationships |
| Lesson | Lessons 6–7 — Converting Length Across Systems; Currency | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.7.AR.3.3 |

## 2. Standards

**MA.7.AR.3.3** Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

**Must.** Convert a length between customary and metric with a printed reference-sheet ratio, either direction, in a chain; convert money with a given rate, both ways. (MA.7.AR.3.3.)

**Must not.** A ratio not on the FAST reference sheet; a real exchange rate the student must know; mass, volume or area (5.08–09).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | Notes II and board 6: the estimate before the arithmetic — a centimeter is shorter than an inch, so fewer inches; a yard is nearly a meter, so about 270 cm |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote on which ratio |
| MA.K12.MTR.3.1 | Complete tasks with mathematical fluency. | boards 5–6: two conversions in a row, each one a form of 1, with the units cancelling on the paper |

## 4. Learning Target and Essential Question

**Learning target.** I can convert a length between the customary and metric systems with a ratio from the reference sheet, and convert money with a given rate.

**Essential question.** 1 in = 2.54 cm and 1 cm = 0.39 in are both on the sheet. How do you know which one to multiply by?

**Building on.** 5.04 — a scale read as a rate; 5.01 — multiplying by a rate; Grade 6 — converting inside one system (12 in = 1 ft).

**Working toward.** 5.08–09 — the same move on mass and volume, and squared for area; 5.10–11 — a conversion inside a rate comparison.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front; say the off-slide sentence out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; due by the Unit 5 test. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Why does multiplying 15 in by 2.54 cm per inch change the unit and not the length? | DOK 2 |
| Jo turned 30 cm into 76.2 in. Without computing, how do you know that is wrong? | DOK 3 |
| Money converts like length, but the ratio is on the problem and not on the sheet. Why? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | $1\text{ in} = 2.54\text{ cm}$ . A ruler is 20 in long. How many centimeters is that? | ***50.8 cm*** | 7.87 — divided by 2.54 [Notes II: centimeters are smaller, so more of them]; 22.54 — added the ratio |
| 2 | $1\text{ cm} = 0.39\text{ in}$ . A phone is 20 cm long. How many inches is that? | ***7.8 in*** | 50.8 — multiplied by 2.54 [Notes II: pick the ratio whose 1 is the unit you have]; 0.78 — place value |
| 3 | $1\text{ mi} = 1.61\text{ km}$ . A race is 5 mi long. How many kilometers is that? | ***8.05 km*** | 3.1 — multiplied by 0.62, the other ratio [Notes II]; 6.61 — added 1.61 |
| 4 | $1\text{ km} = 0.62\text{ mi}$ . A trail is 5 km long. How many miles is that? | ***D — 3.1 mi*** | (A) 5 + 0.62: the additive error [7.AR.3.3: Math Nation's named misconception]; (B) place value: 5 × 0.62 is 3.1, not 0.31 [4.03]; (C) × 1.61, the ratio for the other direction [B1G-M 7.AR.3.3: the inverted ratio; Notes II] |
| 5 | $1\text{ ft} = 12\text{ in}$ and $1\text{ in} = 2.54\text{ cm}$ . A door is 6 ft tall. How many centimeters is that? | ***182.88 cm*** | 15.24 — 6 × 2.54, the feet treated as inches [Notes I: the unit you have must match the ratio's 1]; 183 — rounded early (accept with the work shown) |
| 6 | $1\text{ yd} = 3\text{ ft}$ , $1\text{ ft} = 12\text{ in}$ , $1\text{ in} = 2.54\text{ cm}$ . A table is 3 yd long. About how many centimeters is that? Estimate, then compute. | ***About 270; exactly 274.32 cm*** | 7.62 — 3 × 2.54, the yards treated as inches [Notes II: estimate first — a 3-yd table is not 8 cm]; 91.44 — stopped at 36 in × 2.54 (one yard) |
| 7 | Rate on Nov 2: $\$1 = 1.32\text{ CAD}$ ; $1\text{ CAD} = 16.2\text{ pesos}$ . Theo changes $200 to CAD, then to pesos. How many pesos does he get? | ***4,276.80 pesos*** | 3,240 — 200 × 16.2, the first hop skipped [Notes III: the rate's 1 must be the currency you have]; 2,454.55 — divided by 1.32 on the first hop |
| 8 | $1\text{ km} = 0.62\text{ mi}$ . Theo, who weighs 110 lb, rides a 12 km trail. How long is the trail in miles? | ***7.44 mi*** | 19.32 — × 1.61 [B1G-M 7.AR.3.3: the inverted ratio]; 49.94 kg — converted the weight instead |
| 9 | $1\text{ in} = 2.54\text{ cm}$ and $1\text{ cm} = 0.39\text{ in}$ . Jo converts 30 cm to inches by multiplying by 2.54 and gets 76.2 in. Name the error and fix it. | ***She used the inches ratio on centimeters. Inches are bigger, so fewer: 11.7 in.*** | 'She should have used 0.39' with no reason — the fix without the error named; 76.2 defended as 'more inches' — the estimate backwards [B1G-M 7.AR.3.3] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** The reference sheet's two conversion lines stay on the board all period. Every board starts by pointing at the ratio whose 1 is the unit you have, then multiplying. The estimate is said aloud before the calculator.

**ELL.** **Customary** is the system with inches and miles; **metric** has centimeters and kilometers. The word **per** in "2.54 cm per inch" is the same per as miles per hour.

**Enrichment.** Board 7 finishes early for some: ask them to run the chain backwards — pesos to Canadian dollars to US dollars — and say which operation each step became.

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the error named (Jo multiplied by the ratio whose 1 is inches, when she had centimeters), the estimate that catches it (inches are longer, so fewer of them), and the fix (30 × 0.39 = 11.7 in). A board that writes the right number with no sentence about the direction has not met the lesson.

**Vocabulary.** **customary**; **metric**; **conversion ratio**; **exchange rate**

**Materials.** Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (they are printed on every item as well). Calculators are allowed on everything and printed on nothing.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 5 test: Convert between customary and metric systems — E8Z; Compare and convert customary units — 4B8; Compare and convert metric units — HW2.

# Converting Mass, Area and Volume

*Teacher's Edition*

## The Period

**MA.7.AR.3.3** Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

**Must** Convert mass and volume with a printed reference-sheet ratio, either direction; convert area by squaring the length ratio (and volume by cubing it, in writing); route fluid ounces through cups. (MA.7.AR.3.3.)

**Must not** A ratio not on the FAST reference sheet (no fl oz → mL, no $ft^2$ → $m^2$ as one number); a density or a weight-from-volume question; the capacity of a solid from its dimensions (Unit 6).

**Target** I can convert mass, volume and area between systems, and square the length ratio for area.

**MTR** **MTR.5.1** — Notes II: yesterday's $k^2$ and today's 144 are the same structure, and the room is asked to say so · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.6.1** — every mass and volume board says its estimate first

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 10 min |
| Example 1 | 8–9 | 4 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 11 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. Mass and volume convert like length: the sheet's ratio, the estimate first.
2. Square units, square the ratio: 1 $ft^2$ is 144 $in^2$, 1 $m^2$ is 10,000 $cm^2$.
3. Fluid ounces go through cups — read the sheet for the path.

## Slide by Slide

1. **Converting Mass, Area and Volume.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is Notes II: a square foot is 12 by 12 inches, 144 of them, not 12. *1 retrieves 5.06–07 — × the ratio whose 1 is inches (yesterday). 2 retrieves 5.02–03 — area × $k^2$; today's 144 is the same idea (earlier this unit). 3 retrieves 4.01 — a signed decimal product (last unit). 4 retrieves MA.6.GR.2.1 — 1 $ft^2$ is a 12 by 12 square; today's Notes II (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 30.48 cm 2. 63 $cm^2$ 3. −10 4. 144 $in^2$
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the two lines.** 1 lb = 0.454 kg is a form of 1 like 1 in = 2.54 cm.
6. **Notes.** 1 ft = 12 in, but a square foot is a 12 by 12 square: 144 $in^2$.
7. **Notes.** The sheet has 1 c = 8 fl oz and 1 c = 0.24 L, and no line from fluid ounces to milliliters — so a fluid ounce reaches liters through cups: a 12-fl-oz can is 12 ÷ 8 = 1.5 c, and 1.5 × 0.24 = 0.36 L.
8. **Example 1 — Pounds to kilograms.** Ask which is heavier, a pound or a kilogram, before anyone multiplies.
9. **Example 1 — Worked.** 31.78 kg.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 40.86 kg — 40.9 to one place.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Square feet to square inches.** Ask how many times the ratio is used before anyone multiplies.
13. **Example 2 — Worked.** 720 $in^2$.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 45 $ft^2$.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** $1\,\text{kg} = 2.204\,\text{lb}$ . A bag of rice is 10 kg. How many pounds is that? → ***22.04 lb*** *4.54 — × 0.454, the other ratio [B1G-M 7.AR.3.3: the inverted ratio]; 12.204 — added*
17. **18–19. Board 2.** $1\,\text{gal} = 3.785\,\text{L}$ . A cooler holds 4 gal. How many liters is that? → ***15.14 L*** *1.06 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 7.785 — added*
18. **20–21. Board 3.** $1\,\text{L} = 0.264\,\text{gal}$ . A tank holds 10 L. How many gallons is that? → ***2.64 gal*** *37.85 — × 3.785 [Notes I: pick the ratio whose 1 is the unit you have]; 0.264 — forgot the 10*
19. **22–23. Board 4.** $1\,\text{yd} = 3\,\text{ft}$ . A patch of lawn is $2\,\text{yd}^2$ in area. How many square feet is that? → ***B — 18 $ft^2$*** *(A) 2 × 6: 'two dimensions, so double the ratio' [5.02–03 Notes II: k × k, not 2k]; (C) $3^2$ alone: the 2 $yd^2$ forgotten — one square yard, not two [4.06: finish the computation]; (D) 2 × 3: the ratio used once [7.AR.3.3: square units need the ratio squared — the unit's own; Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** $1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$. A bottle holds 16 fl oz. How many liters is that? → ***0.48 L*** *3.84 — 16 × 0.24, the fluid ounces treated as cups [Notes III: the sheet's path is fl oz → c → L]; 453.6 — used 28.35, the weight ounce [vocabulary: oz and fl oz are two units]*

**26–27. Board 6.** $1\,\text{m} = 100\,\text{cm}$. How many square centimeters is $1\,\text{m}^2$? → ***C — 10,000 cm²*** *(A) 10 × 100: the ratio once, then a guess upward [7.AR.3.3: square units need the ratio squared — the unit's own]; (B) 2 × 100: 'two dimensions, so double it' [5.02–03 Notes II]; (D) the length ratio handed in for an area [7.AR.3.3: square units need the ratio squared — the unit's own; Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** $1\,\text{kg} = 2.204\,\text{lb}$. A 5-lb bag of flour and a 2.5-kg bag of sugar. Which is heavier, and by how much? → ***The sugar, by 0.51 lb*** *'the flour, because 5 > 2.5' — the numbers compared, not the weights [MTR.6.1: units first]; 'the flour, by 2.5 lb' — 5 − 2.5 in two units*

**30–31. Board 8.** $1\,\text{gal} = 3.785\,\text{L}$. A 20-gal fish tank, bought for $45, is being filled. How many liters does it hold? → ***75.7 L*** *The $45 is not needed. 5.28 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 170.33 — the price used*

**32–33. Board 9.** $1\,\text{ft} = 12\,\text{in}$. Priya says $2\,\text{ft}^3$ is $24\,\text{in}^3$, because $2 \times 12 = 24$. Name the error and fix it. → ***She used the ratio once. A cubic foot is 12 × 12 × 12 = 1,728 in³, so 2 ft³ = 3,456 in³.*** *288 in³ — the ratio squared, today's rule applied to a cube [Notes III: three lengths, three times]; 'it should be 24 ft' — the unit changed instead of the number*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–3 take one ratio each direction — mass, then volume both ways; 4 asks a square unit as a vote; 5 chains fluid ounces through cups. The pattern breaks at 6, where the familiar 100 is wrong. 7 converts and then compares, 8 is the extra number, 9 the cube in writing. The independent set walks the same path: mass both ways, volume, a square unit, fluid ounces through cups, square meters.

## Misconceptions to Watch

**Board 1.** 4.54 — × 0.454, the other ratio [B1G-M 7.AR.3.3: the inverted ratio]

**Board 2.** 1.06 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]

**Board 3.** 37.85 — × 3.785 [Notes I: pick the ratio whose 1 is the unit you have]

**Board 4.** 2 × 6: 'two dimensions, so double the ratio' [5.02–03 Notes II: k × k, not 2k]

**Board 5.** 3.84 — 16 × 0.24, the fluid ounces treated as cups [Notes III: the sheet's path is fl oz → c → L]

**Board 6.** 10 × 100: the ratio once, then a guess upward [7.AR.3.3: square units need the ratio squared — the unit's own]

**Board 7.** 'the flour, because 5 > 2.5' — the numbers compared, not the weights [MTR.6.1: units first]

**Board 8.** 5.28 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]

**Board 9.** 288 $\text{in}^3$ — the ratio squared, today's rule applied to a cube [Notes III: three lengths, three times]

**Materials** Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (printed on every item as well); a 12 by 12 grid under the

document camera for Notes II. The Unit Review paper to send home. Calculators are allowed on everything and printed on nothing.

# Converting Mass, Area and Volume

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 5 — Multi-Step Proportional Relationships |
| Lesson | Lessons 8–9 — Converting Mass, Area and Volume | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.7.AR.3.3 |

## 2. Standards

**MA.7.AR.3.3** Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

**Must.** Convert mass and volume with a printed reference-sheet ratio, either direction; convert area by squaring the length ratio (and volume by cubing it, in writing); route fluid ounces through cups. (MA.7.AR.3.3.)

**Must not.** A ratio not on the FAST reference sheet (no fl oz → mL, no $ft^2$ → $m^2$ as one number); a density or a weight-from-volume question; the capacity of a solid from its dimensions (Unit 6).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts. | Notes II: yesterday's $k^2$ and today's 144 are the same structure, and the room is asked to say so |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | every mass and volume board says its estimate first — fewer kilograms than pounds, more liters than gallons |

## 4. Learning Target and Essential Question

**Learning target.** I can convert mass, volume and area between systems, and square the length ratio for area.

**Essential question.** 1 ft = 12 in. Why is 1 $ft^2$ not 12 $in^2$?

**Building on.** 5.06–07 — multiply by the sheet's ratio, estimate first; 5.02–03 — area is two lengths, so k is used twice; Grade 6 — 1 $ft^2$ is a 12 by 12 square.

**Working toward.** 5.10–11 — a conversion inside a multi-step problem; Unit 6 — surface area and volume in whichever unit the problem names.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front; say the off-slide sentence out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; due by the Unit 5 test. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| 1 m = 100 cm. Why is 1 $m^2$ = 10,000 $cm^2$ and not 100? | DOK 3 |
| A 5-lb bag and a 2-kg bag: which is heavier, and what has to happen before you can say? | DOK 2 |
| Why do fluid ounces go through cups to reach liters on this reference sheet? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | $1\,\text{kg} = 2.204\,\text{lb}$. A bag of rice is 10 kg. How many pounds is that? | ***22.04 lb*** | 4.54 — × 0.454, the other ratio [B1G-M 7.AR.3.3: the inverted ratio]; 12.204 — added |
| 2 | $1\,\text{gal} = 3.785\,\text{L}$. A cooler holds 4 gal. How many liters is that? | ***15.14 L*** | 1.06 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 7.785 — added |
| 3 | $1\,\text{L} = 0.264\,\text{gal}$. A tank holds 10 L. How many gallons is that? | ***2.64 gal*** | 37.85 — × 3.785 [Notes I: pick the ratio whose 1 is the unit you have]; 0.264 — forgot the 10 |
| 4 | $1\,\text{yd} = 3\,\text{ft}$. A patch of lawn is $2\,\text{yd}^2$ in area. How many square feet is that? | ***B — 18 ft²*** | (A) 2 × 6: 'two dimensions, so double the ratio' [5.02–03 Notes II: k × k, not 2k]; (C) $3^2$ alone: the 2 $\text{yd}^2$ forgotten — one square yard, not two [4.06: finish the computation]; (D) 2 × 3: the ratio used once [7.AR.3.3: square units need the ratio squared — the unit's own; Notes II] |
| 5 | $1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$. A bottle holds 16 fl oz. How many liters is that? | ***0.48 L*** | 3.84 — 16 × 0.24, the fluid ounces treated as cups [Notes III: the sheet's path is fl oz → c → L]; 453.6 — used 28.35, the weight ounce [vocabulary: oz and fl oz are two units] |
| 6 | $1\,\text{m} = 100\,\text{cm}$. How many square centimeters is $1\,\text{m}^2$? | ***C — 10,000 cm²*** | (A) 10 × 100: the ratio once, then a guess upward [7.AR.3.3: square units need the ratio squared — the unit's own]; (B) 2 × 100: 'two dimensions, so double it' [5.02–03 Notes II]; (D) the length ratio handed in for an area [7.AR.3.3: square units need the ratio squared — the unit's own; Notes II] |
| 7 | $1\,\text{kg} = 2.204\,\text{lb}$. A 5-lb bag of flour and a 2.5-kg bag of sugar. Which is heavier, and by how much? | ***The sugar, by 0.51 lb*** | 'the flour, because 5 > 2.5' — the numbers compared, not the weights [MTR.6.1: units first]; 'the flour, by 2.5 lb' — 5 − 2.5 in two units |
| 8 | $1\,\text{gal} = 3.785\,\text{L}$. A 20-gal fish tank, bought for $45, is being filled. How many liters does it hold? | ***75.7 L*** | 5.28 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 170.33 — the price used |
| 9 | $1\,\text{ft} = 12\,\text{in}$. Priya says $2\,\text{ft}^3$ is $24\,\text{in}^3$, because $2 \times 12 = 24$. Name the error and fix it. | ***She used the ratio once. A cubic foot is 12 × 12 × 12 = 1,728 in³, so 2 ft³ = 3,456 in³.*** | 288 $\text{in}^3$ — the ratio squared, today's rule applied to a cube [Notes III: three lengths, three times]; 'it should be 24 ft' — the unit changed instead of the number |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** Three lines on the board all period: length — the ratio once; area — twice; volume — three times. Every board is answered by pointing at one line first. A 12 by 12 grid stays up beside them.

**ELL.** **Mass** and **weight** are one word here; **ounce** and **fluid ounce** are two different things — say "weight ounce" and "liquid ounce" aloud. **Square** means two lengths; **cubic** means three.

**Enrichment.** Board 9 finishes early for some: ask for 1 $\text{m}^3$ in $\text{cm}^3$ (1,000,000) and whether 1 L = 1,000 $\text{cm}^3$ makes sense from it (1 L = 1,000 mL; a liter is a 10-cm cube).

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the error named (Priya used the ratio once, and a cubic foot is three lengths), and the fix ($12 \times 12 \times 12 = 1{,}728$ $in^3$ per $ft^3$, so 2 $ft^3$ = 3,456 $in^3$). A board that writes 3,456 with no sentence about three lengths has named the number, not the reason.

**Vocabulary.** **mass and weight**; **ounce (oz)**; **fluid ounce (fl oz)**; **square units**; **cubic units**

**Materials.** Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (printed on every item as well); a 12 by 12 grid under the document camera for Notes II. The Unit Review paper to send home. Calculators are allowed on everything and printed on nothing.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 5 test: Convert between customary and metric systems — E8Z; Compare and convert customary units — 4B8; Compare and convert metric units — HW2.

# Multi-Step Proportion Problems

*Teacher's Edition*

## The Period

**MA.7.AR.4.5** Solve real-world problems involving proportional relationships.

**Must** Solve a two- or three-step proportional problem (a rate used twice; a recipe scaled and checked; a part-to-whole mixture), and compare two rates after putting them in one unit with a printed reference-sheet ratio. (MA.7.AR.4.5, with MA.7.AR.3.3 inside it.)

**Must not** A graph, a table of a proportional relationship, or y = px named as such (MA.7.AR.4.1–4.4, Unit 7); a ratio not on the FAST sheet; a percent problem (Unit 8).

**Target** I can solve a problem that takes two or more proportional steps, including a unit conversion, and compare two rates given in different units.

**MTR** **MTR.1.1** — Notes I and board 4: a problem whose first answer (miles per tank) is not the answer, and the room keeps going · **MTR.6.1** — Notes II and board 6: 'which is faster?' is refused until the units match; the answer is checked against the estimate that a mile is longer · **MTR.7.1** — boards 8–9: a plan's area in square meters and a choice of car

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 10 min |
| Example 1 | 8–9 | 4 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 11 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. A multi-step problem is one unit rate used again — name what each step finds.
2. Same unit first; then compare.
3. A recipe's k comes from the yields; then every ingredient, then the pantry.

## Slide by Slide

1. **Multi-Step Proportion Problems.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is the unit rate every board today starts from. *1 retrieves 5.08–09 — × the ratio whose 1 is kilograms (yesterday). 2 retrieves 5.04 — drawing × rate (earlier this unit). 3 retrieves 4.04 — a two-step with decimals (last unit). 4 retrieves MA.6.AR.3.2 — a unit rate; today every board starts from one (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 8.816 lb 2. 150 km 3. 9 4. 60 miles per hour
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the two lines.** A car gets 23 miles per gallon and its tank holds 17.5 gal: 23 × 17.5 = 402.5 mi on a tank.
6. **Notes.** 100 km/h against 65 mi/h: nothing compares until the units match.
7. **Notes.** 12 cookies take 1½ cups of flour.
8. **Example 1 — Two rates in different units.** Ask which unit the comparison will be made in before anyone computes.
9. **Example 1 — Worked.** A: 8.05 km in ⅔ h is 12.075 km/h; B: 9 km in ¾ h is 12 km/h.
10. **Your Turn — Same steps, your numbers. Boards up when done.** A: 6.44 km in ½ h is 12.88 km/h; B: 7 km in 7⁄12 h is 12 km/h.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A recipe scaled, then checked.** Ask what k is before anyone touches the rice.
13. **Example 2 — Worked.** k = 2.5, so 6¼ cups; 5 cups is 2 recipes, 16 servings.
14. **Your Turn — Same steps, your numbers. Boards up when done.** k = 2, 3 cups; 4½ cups is 3 recipes, 18 servings.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

**16–17. Board 1.** A car gets 28 miles per gallon. Its tank holds 15 gal. How far does it go on a full tank? → ***420 mi*** *1.87 — 28 ÷ 15 [Notes I: miles per gallon × gallons gives miles]; 43 — added*

**18–19. Board 2.** A printer prints 35 pages per minute. A job is 1,400 pages. How long does the job take? → ***40 min*** *49,000 — multiplied [Notes I: name what the step finds — pages × pages/min is not minutes]; 400 — place value*

**20–21. Board 3.** Mia earns $14.50 per hour at a shop 2 mi from home, 6 h a day for 5 days. How much does she earn? → ***$435*** *The 2 mi is not needed. 87 — one day's pay [Notes I: the question asks for five days]; 2,175 — 435 × 5, the days counted twice*

**22–23. Board 4.** A car gets 30 miles per gallon with a 12-gal tank. It makes a 900-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) → ***D — 3 tanks*** *(A) 2.5 rounded down: half a tank of road left undriven [Notes I: a part of a tank is still a tank to buy]; (B) 900 ÷ 12: the gallons used as miles per tank [Notes I: the rate first]; (C) 900 ÷ 36: the two numbers added before dividing [7.AR.4.5: the steps not named — Notes I] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** A cereal has 12 g of protein per 100 g. How many grams of protein in 1 kg? → ***120 g*** *12 — the rate copied with the wrong unit [Notes I: per 100 g, and a kilogram is ten of those]; 1,200 — × 100*

**26–27. Board 6.** $1\,\text{mi} = 1.61\,\text{km}$. Cyclist A rides 12 km in 50 min; cyclist B rides 7 mi in 45 min. Who is faster? → ***C — cyclist B*** *(A) 12 km and 12 km/h confused: a distance read as a speed [7.AR.4.5: a rate is per unit of time — Notes II]; (B) 12 > 7, the distances compared in two units [Notes II: same unit first]; (D) 11.27 is B's distance in km, not a speed — the time was never used [7.AR.4.5: a rate is per unit of time — Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** Paint is mixed white : blue : grey = $2:4:5$. A batch uses 8 gal of white. How many gallons is the whole batch? → ***44 gal*** *20 — the grey (5 × 4) and nothing more [Notes III: the whole is all the parts]; 32 — 8 × k, the white scaled again*

**30–31. Board 8.** A plan is drawn at $1\,\text{in} : 4\,\text{ft}$; a room is 2 in by 2.5 in on it. $1\,\text{ft} = 0.305\,\text{m}$. Find the room's area in square meters. → ***About 7.44 m²*** *24.4 — 80 ft² × 0.305, the ratio once on an area [5.08–09 Notes II]; 20 — 2 × 2.5 × 4, the drawing's area scaled once [5.04 board 6]*

**32–33. Board 9.** $1\,\text{mi} = 1.61\,\text{km}$. A 300-km drive. Car A averages 50 mi/h; car B averages 75 km/h. Which car gets there first, and by how long? Explain. → ***Car A: 80.5 km/h against 75, so about 3.73 h against 4 h — about 16 minutes sooner.*** *'Car B, 75 > 50' — numbers compared in two units [Notes II: same unit first]; 'Car A, 80.5 > 75' with no times — the decision without the 'by how long'*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–3 use one rate once, backwards, and on a total, the numbers growing; 4 asks the second step as a vote; 5 changes the unit inside the rate. The pattern breaks at 6, where the comparison cannot be made until the units match. 7 is part-to-whole; 8 and 9 are the test's transfer shapes rehearsed. The independent set walks the same path: a range, the tanks, a recipe, a comparison in one unit, a mixture, a plan's area in square meters.

## Misconceptions to Watch

**Board 1.** 1.87 — 28 ÷ 15 [Notes I: miles per gallon × gallons gives miles]

**Board 2.** 49,000 — multiplied [Notes I: name what the step finds — pages × pages/min is not minutes]

**Board 3.** 87 — one day's pay [Notes I: the question asks for five days]

**Board 4.** 2.5 rounded down: half a tank of road left undriven [Notes I: a part of a tank is still a tank to buy]

**Board 5.** 12 — the rate copied with the wrong unit [Notes I: per 100 g, and a kilogram is ten of those]

**Board 6.** 12 km and 12 km/h confused: a distance read as a speed [7.AR.4.5: a rate is per unit of time — Notes II]

**Board 7.** 20 — the grey (5 × 4) and nothing more [Notes III: the whole is all the parts]

**Board 8.** 24.4 — 80 ft² × 0.305, the ratio once on an area [5.08–09 Notes II]

**Board 9.** 'Car B, 75 > 50' — numbers compared in two units [Notes II: same unit first]

**Materials** Whiteboards and markers. The FAST reference sheet's conversion lines on the board (printed on every item as well). The Unit Review key to post. Calculators are allowed on everything and printed on nothing. 43 minutes: the timings below are the short ones.

# Multi-Step Proportion Problems

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 5 — Multi-Step Proportional Relationships |
| Lesson | Lessons 10–11 — Multi-Step Proportion Problems | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.7.AR.4.5 |

## 2. Standards

**MA.7.AR.4.5** Solve real-world problems involving proportional relationships.

**Must.** Solve a two- or three-step proportional problem (a rate used twice; a recipe scaled and checked; a part-to-whole mixture), and compare two rates after putting them in one unit with a printed reference-sheet ratio. (MA.7.AR.4.5, with MA.7.AR.3.3 inside it.)

**Must not.** A graph, a table of a proportional relationship, or y = px named as such (MA.7.AR.4.1–4.4, Unit 7); a ratio not on the FAST sheet; a percent problem (Unit 8).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.1.1 | Actively participate in effortful learning both individually and collectively. | Notes I and board 4: a problem whose first answer (miles per tank) is not the answer, and the room keeps going |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | Notes II and board 6: 'which is faster?' is refused until the units match; the answer is checked against the estimate that a mile is longer |
| MA.K12.MTR.7.1 | Apply mathematics to real-world contexts. | boards 8–9: a plan's area in square meters and a choice of car — the unit's whole toolkit on one real question |

## 4. Learning Target and Essential Question

**Learning target.** I can solve a problem that takes two or more proportional steps, including a unit conversion, and compare two rates given in different units.

**Essential question.** A car gets 100 km/h and another does 65 mi/h. Why can't you say which is faster yet?

**Building on.** 5.06–09 — the sheet's ratios; 5.04 — a scale, then a speed: two rates in one problem; Grade 6 — a unit rate.

**Working toward.** Unit 7 — the constant of proportionality as a graph and an equation; the Unit 5 test's two transfer items, which are boards 8 and 9 in a third context.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front; say the off-slide sentence out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; due by the Unit 5 test. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| A car goes 402.5 mi on a tank. How many tanks does a 1,207-mi trip take, and why is the answer a whole number? | DOK 2 |
| Two rates are 12 km in 50 min and 7 mi in 45 min. Why does 12 > 7 tell you nothing yet? | DOK 3 |
| Paint is mixed 2 : 4 : 5. If you know the white, how do you find the whole batch — and why is that a scale factor? | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | A car gets 28 miles per gallon. Its tank holds 15 gal. How far does it go on a full tank? | ***420 mi*** | 1.87 — 28 ÷ 15 [Notes I: miles per gallon × gallons gives miles]; 43 — added |
| 2 | A printer prints 35 pages per minute. A job is 1,400 pages. How long does the job take? | ***40 min*** | 49,000 — multiplied [Notes I: name what the step finds — pages × pages/min is not minutes]; 400 — place value |
| 3 | Mia earns $14.50 per hour at a shop 2 mi from home, 6 h a day for 5 days. How much does she earn? | ***$435*** | 87 — one day's pay [Notes I: the question asks for five days]; 2,175 — 435 × 5, the days counted twice |
| 4 | A car gets 30 miles per gallon with a 12-gal tank. It makes a 900-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) | ***D — 3 tanks*** | (A) 2.5 rounded down: half a tank of road left undriven [Notes I: a part of a tank is still a tank to buy]; (B) 900 ÷ 12: the gallons used as miles per tank [Notes I: the rate first]; (C) 900 ÷ 36: the two numbers added before dividing [7.AR.4.5: the steps not named — Notes I] |
| 5 | A cereal has 12 g of protein per 100 g. How many grams of protein in 1 kg? | ***120 g*** | 12 — the rate copied with the wrong unit [Notes I: per 100 g, and a kilogram is ten of those]; 1,200 — × 100 |
| 6 | $1\,\text{mi} = 1.61\,\text{km}$ . Cyclist A rides 12 km in 50 min; cyclist B rides 7 mi in 45 min. Who is faster? | ***C — cyclist B*** | (A) 12 km and 12 km/h confused: a distance read as a speed [7.AR.4.5: a rate is per unit of time — Notes II]; (B) 12 > 7, the distances compared in two units [Notes II: same unit first]; (D) 11.27 is B's distance in km, not a speed — the time was never used [7.AR.4.5: a rate is per unit of time — Notes II] |
| 7 | Paint is mixed white : blue : grey = $2:4:5$ . A batch uses 8 gal of white. How many gallons is the whole batch? | ***44 gal*** | 20 — the grey (5 × 4) and nothing more [Notes III: the whole is all the parts]; 32 — 8 × k, the white scaled again |
| 8 | A plan is drawn at $1\,\text{in} : 4\,\text{ft}$ ; a room is 2 in by 2.5 in on it. $1\,\text{ft} = 0.305\,\text{m}$ . Find the room's area in square meters. | ***About 7.44 m²*** | 24.4 — 80 ft² × 0.305, the ratio once on an area [5.08–09 Notes II]; 20 — 2 × 2.5 × 4, the drawing's area scaled once [5.04 board 6] |
| 9 | $1\,\text{mi} = 1.61\,\text{km}$ . A 300-km drive. Car A averages 50 mi/h; car B averages 75 km/h. Which car gets there first, and by how long? Explain. | ***Car A: 80.5 km/h against 75, so about 3.73 h against 4 h — about 16 minutes sooner.*** | 'Car B, 75 > 50' — numbers compared in two units [Notes II: same unit first]; 'Car A, 80.5 > 75' with no times — the decision without the 'by how long' |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** Every board starts with the unit rate written as 'per 1' before anything is multiplied. For a comparison, a two-column table — one unit across the top — stays on the board, and nothing is compared until both columns are filled.

**ELL.** **Per** means for each one; **rate** is a per-sentence (miles per gallon, pages per minute). **Tank** on a board means a full tank of fuel.

**Enrichment.** Board 7 finishes early for some: ask for the batch if the grey were 25 gal instead ($k = 5$, 55 gal), and why the whole is always $11 \times k$.

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the two speeds in one unit (50 mi/h is 80.5 km/h), the comparison, and the decision in a sentence with its numbers. A board that picked a car without converting has compared numbers, not speeds.

**Vocabulary.** **unit rate**; **proportional**; **per**

**Materials.** Whiteboards and markers. The FAST reference sheet's conversion lines on the board (printed on every item as well). The Unit Review key to post. Calculators are allowed on everything and printed on nothing. 43 minutes: the timings below are the short ones.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 5 test: Compare rates: word problems — VVS; Multi-step problems with customary or metric unit conversions — UCT.

GRADE 7 ACCELERATED · UNIT 5

# Unit 5 Review

*Multi-Step Proportional Relationships*

Name_____
Date_____
Period_____

*Show your work. Circle your final answer.*

*HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it one section at a time. If you miss two or more in a section, that section is where to spend your study time. The reference sheet has everything you need for each section.*

## Scale Drawings

*Lesson 5.01 · 5 questions*

**1.** A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

**2.** A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

**3.** A scale copy is made at $k = 1.5$. A side of the original is 6 cm. Find the matching side of the copy.

**4.** A drawing has a 65° angle. Its scale copy is made at $k = 2$. What is the matching angle in the copy?

- ○ A. 32.5°
- ○ B. 67°
- ○ C. 130°
- ○ D. 65°

**5.** A map is drawn at $1\,\text{cm} : 30\,\text{km}$. Two towns are 4.5 cm apart on it. How far apart are they?

## Scale Factor: Perimeter and Area

*Lessons 5.02–5.03 · 6 questions*

**6.** A figure with perimeter 16 cm is scaled by $k = 2.5$. Find the copy's perimeter.

**7.** A figure with area $9\,\text{cm}^2$ is scaled by $k = 2.5$. Find the copy's area.

**8.** A 4 cm by 6 cm rectangle is scaled by $k = \frac{1}{2}$.

**a.** Find the copy's perimeter.

**b.** Find the copy's area.

**9.** A figure has area $10\,\text{cm}^2$ and its scale copy has area $160\,\text{cm}^2$. Find $k$.

**10.** A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

**11.** Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

## Real-World Scale Drawings

*Lesson 5.04 · 4 questions*

**12.** A plan is drawn at $\frac{1}{2}\,\text{in} : 1\,\text{ft}$. A wall is 7.5 in long on it. Find the real length.

**13.** A plan is drawn at $1\,\text{in} : 6\,\text{ft}$. A real wall is 21 ft long. Find its length on the plan.

**14.** A model boat is built at $1 : 25$ and is 32 cm long. Find the real boat's length in meters.

**15.** Every part below is about one floor plan drawn at $\frac{1}{4}\,\text{in} : 1\,\text{ft}$, on which a room is 3 in by 4 in. $1\,\text{ft} = 0.305\,\text{m}$.

**a.** Find the real length of the wall drawn 3 in long.

**b.** Find the room's real area.

**c.** Write the scale factor from the drawing to the room as a single number.

**d.** Find the room's real perimeter in meters.

## Converting Length Across Systems; Currency

*Lessons 5.06–5.07 · 6 questions*

**16.** $1\,\text{in} = 2.54\,\text{cm}$. How many centimeters is 14 in?

**17.** $1\,\text{cm} = 0.39\,\text{in}$. How many inches is 120 cm?

**18.** $1\,\text{km} = 0.62\,\text{mi}$. How many miles is 30 km?

**19.** $1\,\text{yd} = 3\,\text{ft}$ , $1\,\text{ft} = 12\,\text{in}$ , $1\,\text{in} = 2.54\,\text{cm}$ . How many centimeters is 6 yd? Estimate first.

**20.** Rate on Nov 3: $1\,\text{US dollar} = 0.92\,\text{euros}$ .

**a.** How many euros is $150?

**b.** How many US dollars is 69 euros?

**21.** $1\,\text{in} = 2.54\,\text{cm}$ ; $1\,\text{cm} = 0.39\,\text{in}$ . Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

## Converting Mass, Area and Volume

*Lessons 5.08–5.09 · 5 questions*

**22.** $1\,\text{lb} = 0.454\,\text{kg}$ . How many kilograms is 80 lb?

**23.** $1\,\text{L} = 0.264\,\text{gal}$ . How many gallons is 40 L?

**24.** $1\text{ ft} = 0.305\text{ m}$ . How many square meters is $8\text{ ft}^2$ ? (Round to two places.)

**25.** $1\text{ c} = 8\text{ fl oz}$ and $1\text{ c} = 0.24\text{ L}$ . How many liters is 48 fl oz?

**26.** $1\text{ m} = 3.28\text{ ft}$ . A desk is $1.5\text{ m}^2$ in area. Which is closest to its area in square feet?

- ○ A. $4.9\text{ ft}^2$
- ○ B. $9.8\text{ ft}^2$
- ○ C. $16.1\text{ ft}^2$
- ○ D. $10.8\text{ ft}^2$

## Multi-Step Proportion Problems

*Lessons 5.10–5.11 · 4 questions*

**27.** A car gets 29 miles per gallon and its tank holds 12 gal.

**a.** How far does it go on a full tank?

**b.** It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

**28.** A recipe for 3 servings uses $1\frac{1}{4}$ cups of rice.

**a.** How much rice for 12 servings?

**b.** With 10 cups on hand, how many servings can be made?

**29.** $1\,\text{km} = 0.62\,\text{mi}$ . A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

**30.** Paint is mixed $1:2:5$ . A batch uses 10 gal of the third color. How many gallons is the whole batch?

GRADE 7 ACCELERATED · UNIT 5

# Unit 5 Review

*Multi-Step Proportional Relationships* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

*HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it one section at a time. If you miss two or more in a section, that section is where to spend your study time. The reference sheet has everything you need for each section.*

## Scale Drawings

*Lesson 5.01 · MA.7.GR.1.5 · 5 questions*

**1.** A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

***k = 1.6***

*copy ÷ original*

**2.** A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

***k = 0.25***

*the copy is smaller, so k is under 1*

**3.** A scale copy is made at $k = 1.5$. A side of the original is 6 cm. Find the matching side of the copy.

***9 cm***

**4.** A drawing has a 65° angle. Its scale copy is made at $k = 2$. What is the matching angle in the copy?

- ○ A. 32.5°
- ○ B. 67°
- ○ C. 130°
- ● **D. 65°**

***D***

**5.** A map is drawn at $1\,\text{cm} : 30\,\text{km}$. Two towns are 4.5 cm apart on it. How far apart are they?

***135 km***

*drawing × rate*

## Scale Factor: Perimeter and Area

*Lessons 5.02–5.03 · MA.7.GR.1.5 · 6 questions*

**6.** A figure with perimeter 16 cm is scaled by $k = 2.5$. Find the copy's perimeter.

***40 cm***

*perimeter × k*

**7.** A figure with area $9\,\text{cm}^2$ is scaled by $k = 2.5$. Find the copy's area.

***56.25 cm²***

*area × $k^2$ = 9 × 6.25*

**8.** A 4 cm by 6 cm rectangle is scaled by $k = \frac{1}{2}$.

**a.** Find the copy's perimeter.

***10 cm***

*20 × ½*

**b.** Find the copy's area.

***6 cm²***

*24 × ¼*

**9.** A figure has area $10\,\text{cm}^2$ and its scale copy has area $160\,\text{cm}^2$. Find $k$.

***k = 4***

*160 ÷ 10 = 16 = $k^2$*

**10.** A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

***9***

*k = 3 from the perimeters; the area ratio is $k^2$*

**11.** Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

***A 2 by 5 (area 10) tripled is 6 by 15 (area 90): nine times, because area is two lengths and each is tripled.***

*the reason and the numbers*

## Real-World Scale Drawings

*Lesson 5.04 · MA.7.GR.1.5 · 4 questions*

**12.** A plan is drawn at $\frac{1}{2}\,\text{in} : 1\,\text{ft}$. A wall is 7.5 in long on it. Find the real length.

***15 ft***

*1 in : 2 ft; 7.5 × 2*

**13.** A plan is drawn at $1\,\text{in} : 6\,\text{ft}$. A real wall is 21 ft long. Find its length on the plan.

***3.5 in***

*real ÷ rate*

**14.** A model boat is built at $1 : 25$ and is 32 cm long. Find the real boat's length in meters.

***8 m***

*32 × 25 = 800 cm*

**15.** **SSDD.** Every part below is about one floor plan drawn at ¼ in : 1 ft, on which a room is 3 in by 4 in. 1 ft = 0.305 m. Same surface, four different questions: a length, an area with the rate used twice, the scale factor as one number, and a length conversion — the four moves of the unit on one room, so that a student learns to read the question and not the picture.

**a.** Find the real length of the wall drawn 3 in long.

***12 ft***

*1 in : 4 ft; 3 × 4*

**b.** Find the room's real area.

***192 ft²***

*12 by 16; the rate used twice*

**c.** Write the scale factor from the drawing to the room as a single number.

***48***

*1 in : 4 ft is 1 in : 48 in*

**d.** Find the room's real perimeter in meters.

***17.08 m***

*2(12 + 16) = 56 ft; 56 × 0.305 (56 ÷ 3.28 = 17.07 by the other ratio). The area in square meters is the test's transfer item and is not rehearsed here (ruling 18).*

## Converting Length Across Systems; Currency

*Lessons 5.06–5.07 · MA.7.AR.3.3 · 6 questions*

**16.** $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 14 in?

***35.56 cm***

**17.** $1\text{ cm} = 0.39\text{ in}$. How many inches is 120 cm?

***46.8 in (47.2 by ÷ 2.54)***

*fewer inches than centimeters*

**18.** $1\text{ km} = 0.62\text{ mi}$. How many miles is 30 km?

***18.6 mi***

**19.** $1\text{ yd} = 3\text{ ft}$, $1\text{ ft} = 12\text{ in}$, $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 6 yd? Estimate first.

***About 540; exactly 548.64 cm***

*216 in × 2.54*

**20.** Rate on Nov 3: $1\text{ US dollar} = 0.92\text{ euros}$ .

**a.** How many euros is $150?

***138 euros***

*× the rate*

**b.** How many US dollars is 69 euros?

***$75***

*÷ the rate*

**21.** $1\text{ in} = 2.54\text{ cm}$ ; $1\text{ cm} = 0.39\text{ in}$ . Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

***Lee used the ratio whose 1 is inches on centimeters. Inches are bigger, so there are fewer: 40 × 0.39 = 15.6 in (or 40 ÷ 2.54 = 15.7).***

*the error named and the estimate*

## Converting Mass, Area and Volume

*Lessons 5.08–5.09 · MA.7.AR.3.3 · 5 questions*

**22.** $1\text{ lb} = 0.454\text{ kg}$ . How many kilograms is 80 lb?

***36.32 kg***

*(80 ÷ 2.204 = 36.30 by the other ratio)*

**23.** $1\text{ L} = 0.264\text{ gal}$ . How many gallons is 40 L?

***10.56 gal***

**24.** $1\text{ ft} = 0.305\text{ m}$ . How many square meters is $8\text{ ft}^2$ ? (Round to two places.)

***About 0.74 $m^2$***

*$8 \times 0.305^2 = 0.7442$: the sheet's length ratio, squared*

**25.** $1\text{ c} = 8\text{ fl oz}$ and $1\text{ c} = 0.24\text{ L}$ . How many liters is 48 fl oz?

***1.44 L***

*6 c × 0.24*

**26.** $1\text{ m} = 3.28\text{ ft}$ . A desk is $1.5\text{ m}^2$ in area. Which is closest to its area in square feet?

- ○ A. $4.9\text{ ft}^2$
- ○ B. $9.8\text{ ft}^2$
- ● **C.** $16.1\text{ ft}^2$
- ○ D. $10.8\text{ ft}^2$

***C***

*$1.5 \times 3.28^2 = 16.1376$.*

## Multi-Step Proportion Problems

*Lessons 5.10–5.11 · MA.7.AR.4.5 · 4 questions*

**27.** A car gets 29 miles per gallon and its tank holds 12 gal.

**a.** How far does it go on a full tank?

***348 mi***

**b.** It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

***3 tanks (800 ÷ 348 ≈ 2.3)***

*a part of a tank is still a tank*

**28.** A recipe for 3 servings uses $1\frac{1}{4}$ cups of rice.

**a.** How much rice for 12 servings?

***5 cups***

*k = 4*

**b.** With 10 cups on hand, how many servings can be made?

***24 servings***

*8 recipes × 3*

**29.** $1\text{ km} = 0.62\text{ mi}$. A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

***The train: 110 km/h = 68.2 mi/h > 65 mi/h***

*same unit first*

**30.** Paint is mixed $1:2:5$. A batch uses 10 gal of the third color. How many gallons is the whole batch?

***16 gal***

*k = 2; 8 parts × 2*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

# Multi-Step Proportional Relationships

*Form A · 24 points*

Name_____
Date_____
Period_____

*Show your work. One point for each lettered part.*

## Scale Drawings and Scale Factors

*10 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** An 8 cm side of a figure is 20 cm in its scale copy. Find the scale factor. *1 pt*

**2.** A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

**3.** A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each side is 3 times as long.
- ☐ B. Each angle is 3 times as large.
- ☐ C. The perimeter is 3 times as long.
- ☐ D. The area is 3 times as large.
- ☐ E. The area is 9 times as large.

**4.** A 3 cm by 4 cm rectangle is scaled by k = 3.

**a.** Find the copy's perimeter. *1 pt*

**b.** Find the copy's area. *1 pt*

**5.** A figure has area 8 $cm^2$. Its scale copy has area 72 $cm^2$. What is the scale factor? *1 pt*

**6.** A floor plan is drawn at ¼ in : 1 ft. A room is 4.5 in by 3.5 in on the plan.

**a.** Find the real length of the wall drawn 4.5 in long. *1 pt*

**b.** Find the room's real area. *1 pt*

**7.** A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

**8.** A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

## Converting Across Systems *8 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☐ A. 70 × 0.39
- ☐ B. 70 × 2.54
- ☐ C. 70 ÷ 2.54
- ☐ D. 70 ÷ 0.39

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

**12.** 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

**13.** 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

**14.** 1 ft = 0.305 m. A patio is 30 $ft^2$ in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 9.15 $m^2$
- ○ B. 0.093 $m^2$
- ○ C. 2.79 $m^2$
- ○ D. 18.3 $m^2$

**15.** Rate on Nov 5: 1 US dollar = 0.92 euros.

**a.** How many euros is $60? *1 pt*

**b.** How many US dollars is 23 euros? *1 pt*

## Multi-Step Proportion Problems

*6 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 26 miles per gallon and its tank holds 15 gal.

**a.** How far does it go on a full tank? *1 pt*

**b.** It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

**17.** A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

**a.** How much rice for 14 servings? *1 pt*

**b.** With 6 cups on hand, how many servings can be made? *1 pt*

**18.** Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

**19.** 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

# Multi-Step Proportional Relationships

*Form A · 24 points* **ANSWER KEY**

Name
Date
Period

*Show your work. One point for each lettered part.*

## Scale Drawings and Scale Factors

*10 points MA.7.GR.1.5*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** An 8 cm side of a figure is 20 cm in its scale copy. Find the scale factor. *1 pt*

***k = 2.5***

*copy ÷ original*

*Scoring: 1 point.*

**2.** A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

***k = 0.4***

*copy ÷ original; the copy is smaller, so k is under 1 — 2.5 is the proportion backwards*

*Scoring: 1 point.*

**3.** A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☒ **A. Each side is 3 times as long.**
- ☐ B. Each angle is 3 times as large.
- ☒ **C. The perimeter is 3 times as long.**
- ☐ D. The area is 3 times as large.
- ☒ **E. The area is 9 times as large.**

***A, C and E***

*All three and nothing else for the point: sides and perimeter × k, area × $k^2$, angles unchanged.*

*Scoring: 1 point.*

**4.** A 3 cm by 4 cm rectangle is scaled by k = 3.

**a.** Find the copy's perimeter. *1 pt*

***42 cm***

*perimeter × k*

**b.** Find the copy's area. *1 pt*

***108 cm²***

*area × k² — 36 multiplied the area by k once*

*Scoring: a 1 · b 1 = 2 points.*

**5.** A figure has area 8 cm². Its scale copy has area 72 cm². What is the scale factor? *1 pt*

***k = 3***

*72 ÷ 8 = 9 is k², so k = √9. A board with 9 handed in the area ratio as k.*

*Scoring: 1 point.*

**6.** A floor plan is drawn at ¼ in : 1 ft. A room is 4.5 in by 3.5 in on the plan.

**a.** Find the real length of the wall drawn 4.5 in long. *1 pt*

***18 ft***

*the scale per 1 inch, then ×: 1 in : 4 ft*

**b.** Find the room's real area. *1 pt*

***252 ft²***

*real dimensions first: 18 by 14 — 15.75 in² × 4 = 63 is the named error*

*Scoring: a 1 · b 1 = 2 points.*

**7.** A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

***4.5 in***

*real ÷ rate*

*Scoring: 1 point.*

**8.** A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

***About 9.30 m² (3.05 m by 3.05 m); 9.29 by the sheet's other ratio***

*TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. 10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft² × 0.305² = 9.3025; by the sheet's 1 m = 3.28 ft, 100 ÷ 3.28² = 9.29 — full credit. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.*

*Scoring: 1 point.*

## Converting Across Systems

*8 points* *MA.7.AR.3.3*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

***22.86 cm (23.1 by ÷ 0.39)***

*× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it*

*Scoring: 1 point.*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☒ **A. 70 × 0.39**
- ☐ B. 70 × 2.54
- ☒ **C. 70 ÷ 2.54**
- ☐ D. 70 ÷ 0.39

***A and C***

*Both sheet routes (27.3 and 27.6) and nothing else for the point.*

*Scoring: 1 point.*

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

***365.76 cm***

*a chain: 144 in, then × 2.54 (144 ÷ 0.39 = 369.2 by the other ratio — full credit). The estimate (a yard is nearly a meter) catches 10.16.*

*Scoring: 1 point.*

**12.** 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

***44.08 lb (44.05 by ÷ 0.454)***

*× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**13.** 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

***18.925 L (18.94 by ÷ 0.264)***

*more liters than gallons; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**14.** 1 ft = 0.305 m. A patio is 30 $ft^2$ in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 9.15 $m^2$
- ○ B. 0.093 $m^2$
- ● **C. 2.79 $m^2$**
- ○ D. 18.3 $m^2$

***C***

*30 × 0.305² = 2.79075 (or 30 ÷ 3.28² = 2.79).*

*Scoring: 1 point.*

**15.** Rate on Nov 5: 1 US dollar = 0.92 euros.

**a.** How many euros is $60? *1 pt*

***55.20 euros***

*× the rate*

**b.** How many US dollars is 23 euros? *1 pt*

***$25***

*÷ the rate; 21.16 multiplied both ways*

*Scoring: a 1 · b 1 = 2 points.*

## Multi-Step Proportion Problems

*6 points MA.7.AR.4.5*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 26 miles per gallon and its tank holds 15 gal.

**a.** How far does it go on a full tank? *1 pt*

***390 mi***

*the rate × the gallons*

**b.** It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

***3 tanks (1,000 ÷ 390 ≈ 2.56)***

*the bracket on the paper settles the rounding; follow-through on part (a)*

*Scoring: a 1 · b 1 = 2 points.*

**17.** A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

**a.** How much rice for 14 servings? *1 pt*

***5¼ cups***

*k = 3.5 from the yields, then the ingredient*

**b.** With 6 cups on hand, how many servings can be made? *1 pt*

***16 servings***

*4 recipes × the yield; 24 scaled the yield by the cups*

*Scoring: a 1 · b 1 = 2 points.*

**18.** Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

***27 cups***

*k = 3 from the orange, then the whole, 9 parts; 6 found the mango and stopped*

*Scoring: 1 point.*

**19.** 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

***Car A, by about 14 minutes: 40 mi/h = 64.4 km/h, so 200 ÷ 64.4 ≈ 3.11 h against 200 ÷ 60 ≈ 3.33 h.***

*TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The point needs the speeds in one unit and the choice with its gap: 0.227 h is 13.7 min — 13 or 14 both earn it (by the sheet's 0.62 route, 124 mi: 3.1 h against 3.33 h, 14 min). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.*

*Scoring: 1 point.*

## Score Tracker

| Benchmark | Questions | Possible | Earned |
|---|---|---|---|
| MA.7.GR.1.5 | 1, 2, 3, 4, 5, 6, 7, 8 | 10 | |
| MA.7.AR.3.3 | 9, 10, 11, 12, 13, 14, 15 | 8 | |
| MA.7.AR.4.5 | 16, 17, 18, 19 | 6 | |
| **Total** | | **24** | |

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

# Multi-Step Proportional Relationships

*Form A · 24 points* **WORKED ANSWERS**

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*Show your work. One point for each lettered part.*

## Scale Drawings and Scale Factors *10 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** An 8 cm side of a figure is 20 cm in its scale copy. Find the scale factor. *1 pt*

***k = 2.5***

*copy ÷ original*

*Scoring: 1 point.*

**2.** A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

***k = 0.4***

*copy ÷ original; the copy is smaller, so k is under 1 — 2.5 is the proportion backwards*

*Scoring: 1 point.*

**3.** A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☒ **A. Each side is 3 times as long.**
- ☐ B. Each angle is 3 times as large.
- ☒ **C. The perimeter is 3 times as long.**
- ☐ D. The area is 3 times as large.
- ☒ **E. The area is 9 times as large.**

***A, C and E***

*All three and nothing else for the point: sides and perimeter × k, area × $k^2$, angles unchanged.*

*Scoring: 1 point.*

**4.** A 3 cm by 4 cm rectangle is scaled by k = 3.

**a.** Find the copy's perimeter. *1 pt*

***42 cm***

*perimeter × k*

**b.** Find the copy's area. *1 pt*

***108 cm²***

*area × k² — 36 multiplied the area by k once*

*Scoring: a 1 · b 1 = 2 points.*

**5.** A figure has area 8 cm². Its scale copy has area 72 cm². What is the scale factor? *1 pt*

***k = 3***

*72 ÷ 8 = 9 is k², so k = √9. A board with 9 handed in the area ratio as k.*

*Scoring: 1 point.*

**6.** A floor plan is drawn at ¼ in : 1 ft. A room is 4.5 in by 3.5 in on the plan.

**a.** Find the real length of the wall drawn 4.5 in long. *1 pt*

***18 ft***

*the scale per 1 inch, then ×: 1 in : 4 ft*

**b.** Find the room's real area. *1 pt*

***252 ft²***

*real dimensions first: 18 by 14 — 15.75 in² × 4 = 63 is the named error*

*Scoring: a 1 · b 1 = 2 points.*

**7.** A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

***4.5 in***

*real ÷ rate*

*Scoring: 1 point.*

**8.** A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

***About 9.30 m² (3.05 m by 3.05 m); 9.29 by the sheet's other ratio***

*10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft² × 0.305² = 9.3025; by the sheet's 1 m = 3.28 ft, 100 ÷ 3.28² = 9.29 — full credit. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.*

*Scoring: 1 point.*

## Converting Across Systems

*8 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

***22.86 cm (23.1 by ÷ 0.39)***

*× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it*

*Scoring: 1 point.*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☒ **A. 70 × 0.39**
- ☐ B. 70 × 2.54
- ☒ **C. 70 ÷ 2.54**
- ☐ D. 70 ÷ 0.39

***A and C***

*Both sheet routes (27.3 and 27.6) and nothing else for the point.*

*Scoring: 1 point.*

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

***365.76 cm***

*a chain: 144 in, then × 2.54 (144 ÷ 0.39 = 369.2 by the other ratio — full credit). The estimate (a yard is nearly a meter) catches 10.16.*

*Scoring: 1 point.*

**12.** 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

***44.08 lb (44.05 by ÷ 0.454)***

*× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**13.** 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

***18.925 L (18.94 by ÷ 0.264)***

*more liters than gallons; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**14.** 1 ft = 0.305 m. A patio is 30 $ft^2$ in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 9.15 $m^2$
- ○ B. 0.093 $m^2$
- ● **C. 2.79 $m^2$**
- ○ D. 18.3 $m^2$

***C***

*30 × 0.305² = 2.79075 (or 30 ÷ 3.28² = 2.79).*

*Scoring: 1 point.*

**15.** Rate on Nov 5: 1 US dollar = 0.92 euros.

**a.** How many euros is $60? *1 pt*

***55.20 euros***

*× the rate*

**b.** How many US dollars is 23 euros? *1 pt*

***$25***

*÷ the rate; 21.16 multiplied both ways*

*Scoring: a 1 · b 1 = 2 points.*

## Multi-Step Proportion Problems *6 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 26 miles per gallon and its tank holds 15 gal.

**a.** How far does it go on a full tank? *1 pt*

***390 mi***

*the rate × the gallons*

**b.** It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

***3 tanks (1,000 ÷ 390 ≈ 2.56)***

*the bracket on the paper settles the rounding; follow-through on part (a)*

*Scoring: a 1 · b 1 = 2 points.*

**17.** A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

**a.** How much rice for 14 servings? *1 pt*

***5¼ cups***

*k = 3.5 from the yields, then the ingredient*

**b.** With 6 cups on hand, how many servings can be made? *1 pt*

***16 servings***

*4 recipes × the yield; 24 scaled the yield by the cups*

*Scoring: a 1 · b 1 = 2 points.*

**18.** Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

***27 cups***

*k = 3 from the orange, then the whole, 9 parts; 6 found the mango and stopped*

*Scoring: 1 point.*

**19.** 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

***Car A, by about 14 minutes: 40 mi/h = 64.4 km/h, so 200 ÷ 64.4 ≈ 3.11 h against 200 ÷ 60 ≈ 3.33 h.***

*The point needs the speeds in one unit and the choice with its gap: 0.227 h is 13.7 min — 13 or 14 both earn it (by the sheet's 0.62 route, 124 mi: 3.1 h against 3.33 h, 14 min). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.*

*Scoring: 1 point.*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

# Multi-Step Proportional Relationships

*Form B · 24 points*

Name_____
Date_____
Period_____

*Show your work. One point for each lettered part.*

## Scale Drawings and Scale Factors

*10 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** A 12 cm side of a figure is 18 cm in its scale copy. Find the scale factor. *1 pt*

**2.** A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

**3.** A figure is scaled by a scale factor of 4. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each angle is 4 times as large.
- ☐ B. The area is 16 times as large.
- ☐ C. Each side is 4 times as long.
- ☐ D. The area is 4 times as large.
- ☐ E. The perimeter is 4 times as long.

**4.** A 2 cm by 5 cm rectangle is scaled by k = 4.

**a.** Find the copy's perimeter. *1 pt*

**b.** Find the copy's area. *1 pt*

**5.** A figure has area 5 $cm^2$. Its scale copy has area 80 $cm^2$. What is the scale factor? *1 pt*

**6.** A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

**a.** Find the real length of the wall drawn 3 in long. *1 pt*

**b.** Find the room's real area. *1 pt*

**7.** A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

**8.** A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

## Converting Across Systems *8 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 11 in? *1 pt*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 90 cm. *1 pt*

- ☐ A. 90 ÷ 0.39
- ☐ B. 90 ÷ 2.54
- ☐ C. 90 × 2.54
- ☐ D. 90 × 0.39

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 5 yd? *1 pt*

**12.** 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

**13.** 1 gal = 3.785 L. How many liters is 9 gal? *1 pt*

**14.** 1 m = 3.28 ft. A floor is 5 $m^2$ in area. Which is closest to its area in square feet? *1 pt*

- ○ A. 53.8 $ft^2$
- ○ B. 16.4 $ft^2$
- ○ C. 10.8 $ft^2$
- ○ D. 32.8 $ft^2$

**15.** Rate on Nov 5: 1 US dollar = 1.35 Canadian dollars.

**a.** How many Canadian dollars is $70? *1 pt*

**b.** How many US dollars is 54 Canadian dollars? *1 pt*

## Multi-Step Proportion Problems

*6 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 22 miles per gallon and its tank holds 20 gal.

**a.** How far does it go on a full tank? *1 pt*

**b.** It makes a 1,500-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

**17.** A recipe for 6 servings uses 2 cups of rice.

**a.** How much rice for 9 servings? *1 pt*

**b.** With 8 cups on hand, how many servings can be made? *1 pt*

**18.** Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second color. How many gallons is the whole batch? *1 pt*

**19.** 1 mi = 1.61 km. A drive is 450 km long. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

# Multi-Step Proportional Relationships

*Form B · 24 points* **ANSWER KEY**

Name
Date
Period

*Show your work. One point for each lettered part.*

## Scale Drawings and Scale Factors

*10 points MA.7.GR.1.5*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** A 12 cm side of a figure is 18 cm in its scale copy. Find the scale factor. *1 pt*

***k = 1.5***

*copy ÷ original*

*Scoring: 1 point.*

**2.** A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

***k = 0.25***

*copy ÷ original; the copy is smaller, so k is under 1 — 4 is the proportion backwards*

*Scoring: 1 point.*

**3.** A figure is scaled by a scale factor of 4. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each angle is 4 times as large.
- ☒ **B. The area is 16 times as large.**
- ☒ **C. Each side is 4 times as long.**
- ☐ D. The area is 4 times as large.
- ☒ **E. The perimeter is 4 times as long.**

***B, C and E***

*All three and nothing else for the point: sides and perimeter × k, area × $k^2$, angles unchanged.*

*Scoring: 1 point.*

**4.** A 2 cm by 5 cm rectangle is scaled by k = 4.

**a.** Find the copy's perimeter. *1 pt*

***56 cm***

*perimeter × k*

**b.** Find the copy's area. *1 pt*

***160 cm²***

*area × k² — 40 multiplied the area by k once*

*Scoring: a 1 · b 1 = 2 points.*

**5.** A figure has area 5 cm². Its scale copy has area 80 cm². What is the scale factor? *1 pt*

***k = 4***

*80 ÷ 5 = 16 is k², so k = √16. A board with 16 handed in the area ratio as k.*

*Scoring: 1 point.*

**6.** A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

**a.** Find the real length of the wall drawn 3 in long. *1 pt*

***6 ft***

*the scale per 1 inch, then ×: 1 in : 2 ft*

**b.** Find the room's real area. *1 pt*

***60 ft²***

*real dimensions first: 6 by 10 — 15 in² × 2 = 30 is the named error*

*Scoring: a 1 · b 1 = 2 points.*

**7.** A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

***5.5 in***

*real ÷ rate*

*Scoring: 1 point.*

**8.** A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

***About 4.47 m² (1.83 m by 2.44 m); 4.46 by the sheet's other ratio***

*TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. 6 ft by 8 ft; 1.83 × 2.44 = 4.4652. The same number by the other route: 48 ft² × 0.305² = 4.4652; by the sheet's 1 m = 3.28 ft, 48 ÷ 3.28² = 4.46 — full credit. 14.64 (48 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.*

*Scoring: 1 point.*

## Converting Across Systems

*8 points* *MA.7.AR.3.3*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 11 in? *1 pt*

***27.94 cm (28.2 by ÷ 0.39)***

*× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it*

*Scoring: 1 point.*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 90 cm. *1 pt*

- ☐ A. 90 ÷ 0.39
- ☒ **B. 90 ÷ 2.54**
- ☐ C. 90 × 2.54
- ☒ **D. 90 × 0.39**

***B and D***

*Both sheet routes (35.1 and 35.4) and nothing else for the point.*

*Scoring: 1 point.*

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 5 yd? *1 pt*

***457.2 cm***

*a chain: 180 in, then × 2.54 (180 ÷ 0.39 = 461.5 by the other ratio — full credit). The estimate catches 12.7.*

*Scoring: 1 point.*

**12.** 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

***27.24 kg (27.22 by ÷ 2.204)***

*× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**13.** 1 gal = 3.785 L. How many liters is 9 gal? *1 pt*

***34.065 L (34.09 by ÷ 0.264)***

*more liters than gallons; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**14.** 1 m = 3.28 ft. A floor is 5 $m^2$ in area. Which is closest to its area in square feet? *1 pt*

- ● **A. 53.8 $ft^2$**
- ○ B. 16.4 $ft^2$
- ○ C. 10.8 $ft^2$
- ○ D. 32.8 $ft^2$

***A***

*5 × $3.28^2$ = 53.79 (or 5 ÷ $0.305^2$ = 53.75).*

*Scoring: 1 point.*

**15.** Rate on Nov 5: 1 US dollar = 1.35 Canadian dollars.

**a.** How many Canadian dollars is $70? *1 pt*

***94.50 Canadian dollars***

*× the rate*

**b.** How many US dollars is 54 Canadian dollars? *1 pt*

***$40***

*÷ the rate; 72.90 multiplied both ways*

*Scoring: a 1 · b 1 = 2 points.*

## Multi-Step Proportion Problems

*6 points MA.7.AR.4.5*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 22 miles per gallon and its tank holds 20 gal.

**a.** How far does it go on a full tank? *1 pt*

***440 mi***

*the rate × the gallons*

**b.** It makes a 1,500-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

***4 tanks (1,500 ÷ 440 ≈ 3.41)***

*the bracket on the paper settles the rounding; follow-through on part (a)*

*Scoring: a 1 · b 1 = 2 points.*

**17.** A recipe for 6 servings uses 2 cups of rice.

**a.** How much rice for 9 servings? *1 pt*

***3 cups***

*k = 1.5 from the yields, then the ingredient*

**b.** With 8 cups on hand, how many servings can be made? *1 pt*

***24 servings***

*4 recipes × the yield; 48 scaled the yield by the cups*

*Scoring: a 1 · b 1 = 2 points.*

**18.** Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second color. How many gallons is the whole batch? *1 pt*

***32 gal***

*k = 4 from the second color, then the whole, 8 parts; 16 found the third color and stopped*

*Scoring: 1 point.*

**19.** 1 mi = 1.61 km. A drive is 450 km long. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

***Truck B, by about 5 minutes: 55 mi/h = 88.55 km/h, so 450 ÷ 90 = 5 h against 450 ÷ 88.55 ≈ 5.08 h.***

*TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The point needs the speeds in one unit and the choice with its gap: 0.082 h is 4.9 min — 5 earns it (by the 0.62 route, 279 mi: 5 h against 5.07 h, 4 min; accept 4–5). 'Truck B, 90 > 55' is right by accident — no credit without the conversion. The surface appears on no review and in no question bank; its type was 5.10 board 9.*

*Scoring: 1 point.*

## Score Tracker

| Benchmark | Questions | Possible | Earned |
|---|---|---|---|
| MA.7.GR.1.5 | 1, 2, 3, 4, 5, 6, 7, 8 | 10 | |
| MA.7.AR.3.3 | 9, 10, 11, 12, 13, 14, 15 | 8 | |
| MA.7.AR.4.5 | 16, 17, 18, 19 | 6 | |
| **Total** | | **24** | |

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

# Multi-Step Proportional Relationships

*Form B · 24 points* **WORKED ANSWERS**

---

*Show your work. One point for each lettered part.*

## Scale Drawings and Scale Factors *10 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** A 12 cm side of a figure is 18 cm in its scale copy. Find the scale factor. *1 pt*

***k = 1.5***

*copy ÷ original*

*Scoring: 1 point.*

**2.** A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

***k = 0.25***

*copy ÷ original; the copy is smaller, so k is under 1 — 4 is the proportion backwards*

*Scoring: 1 point.*

**3.** A figure is scaled by a scale factor of 4. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each angle is 4 times as large.
- ☒ **B. The area is 16 times as large.**
- ☒ **C. Each side is 4 times as long.**
- ☐ D. The area is 4 times as large.
- ☒ **E. The perimeter is 4 times as long.**

***B, C and E***

*All three and nothing else for the point: sides and perimeter × k, area × $k^2$, angles unchanged.*

*Scoring: 1 point.*

**4.** A 2 cm by 5 cm rectangle is scaled by k = 4.

**a.** Find the copy's perimeter. *1 pt*

***56 cm***

*perimeter × k*

**b.** Find the copy's area. *1 pt*

***160 cm²***

*area × k² — 40 multiplied the area by k once*

*Scoring: a 1 · b 1 = 2 points.*

**5.** A figure has area 5 cm². Its scale copy has area 80 cm². What is the scale factor? *1 pt*

***k = 4***

*80 ÷ 5 = 16 is k², so k = √16. A board with 16 handed in the area ratio as k.*

*Scoring: 1 point.*

**6.** A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

**a.** Find the real length of the wall drawn 3 in long. *1 pt*

***6 ft***

*the scale per 1 inch, then ×: 1 in : 2 ft*

**b.** Find the room's real area. *1 pt*

***60 ft²***

*real dimensions first: 6 by 10 — 15 in² × 2 = 30 is the named error*

*Scoring: a 1 · b 1 = 2 points.*

**7.** A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

***5.5 in***

*real ÷ rate*

*Scoring: 1 point.*

**8.** A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

***About 4.47 m² (1.83 m by 2.44 m); 4.46 by the sheet's other ratio***

*6 ft by 8 ft; 1.83 × 2.44 = 4.4652. The same number by the other route: 48 ft² × 0.305² = 4.4652; by the sheet's 1 m = 3.28 ft, 48 ÷ 3.28² = 4.46 — full credit. 14.64 (48 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.*

*Scoring: 1 point.*

## Converting Across Systems

*8 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 11 in? *1 pt*

***27.94 cm (28.2 by ÷ 0.39)***

*× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it*

*Scoring: 1 point.*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 90 cm. *1 pt*

- ☐ A. 90 ÷ 0.39
- ☒ **B. 90 ÷ 2.54**
- ☐ C. 90 × 2.54
- ☒ **D. 90 × 0.39**

***B and D***

*Both sheet routes (35.1 and 35.4) and nothing else for the point.*

*Scoring: 1 point.*

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 5 yd? *1 pt*

***457.2 cm***

*a chain: 180 in, then × 2.54 (180 ÷ 0.39 = 461.5 by the other ratio — full credit). The estimate catches 12.7.*

*Scoring: 1 point.*

**12.** 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

***27.24 kg (27.22 by ÷ 2.204)***

*× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**13.** 1 gal = 3.785 L. How many liters is 9 gal? *1 pt*

***34.065 L (34.09 by ÷ 0.264)***

*more liters than gallons; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**14.** 1 m = 3.28 ft. A floor is 5 $m^2$ in area. Which is closest to its area in square feet? *1 pt*

- ● **A. 53.8 $ft^2$**
- ○ B. 16.4 $ft^2$
- ○ C. 10.8 $ft^2$
- ○ D. 32.8 $ft^2$

***A***

*$5 \times 3.28^2 = 53.79$ (or $5 \div 0.305^2 = 53.75$).*

*Scoring: 1 point.*

**15.** Rate on Nov 5: 1 US dollar = 1.35 Canadian dollars.

**a.** How many Canadian dollars is $70? *1 pt*

***94.50 Canadian dollars***

*× the rate*

**b.** How many US dollars is 54 Canadian dollars? *1 pt*

***$40***

*÷ the rate; 72.90 multiplied both ways*

*Scoring: a 1 · b 1 = 2 points.*

## Multi-Step Proportion Problems *6 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 22 miles per gallon and its tank holds 20 gal.

**a.** How far does it go on a full tank? *1 pt*

***440 mi***

*the rate × the gallons*

**b.** It makes a 1,500-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

***4 tanks (1,500 ÷ 440 ≈ 3.41)***

*the bracket on the paper settles the rounding; follow-through on part (a)*

*Scoring: a 1 · b 1 = 2 points.*

**17.** A recipe for 6 servings uses 2 cups of rice.

**a.** How much rice for 9 servings? *1 pt*

***3 cups***

*k = 1.5 from the yields, then the ingredient*

**b.** With 8 cups on hand, how many servings can be made? *1 pt*

***24 servings***

*4 recipes × the yield; 48 scaled the yield by the cups*

*Scoring: a 1 · b 1 = 2 points.*

**18.** Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second color. How many gallons is the whole batch? *1 pt*

***32 gal***

*k = 4 from the second color, then the whole, 8 parts; 16 found the third color and stopped*

*Scoring: 1 point.*

**19.** 1 mi = 1.61 km. A drive is 450 km long. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

***Truck B, by about 5 minutes: 55 mi/h = 88.55 km/h, so 450 ÷ 90 = 5 h against 450 ÷ 88.55 ≈ 5.08 h.***

*The point needs the speeds in one unit and the choice with its gap: 0.082 h is 4.9 min — 5 earns it (by the 0.62 route, 279 mi: 5 h against 5.07 h, 4 min; accept 4–5). 'Truck B, 90 > 55' is right by accident — no credit without the conversion. The surface appears on no review and in no question bank; its type was 5.10 board 9.*

*Scoring: 1 point.*

GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

# Multi-Step Proportional Relationships

*24 points*

Name_____
Date_____
Period_____

*Show your work. One point for each lettered part.*

## Scale Drawings and Scale Factors

*10 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

**2.** A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

**3.** A figure is scaled by a scale factor of 2. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. The area is 2 times as large.
- ☐ B. Each side is 2 times as long.
- ☐ C. The perimeter is 2 times as long.
- ☐ D. The area is 4 times as large.
- ☐ E. Each angle is 2 times as large.

**4.** A 3 cm by 5 cm rectangle is scaled by k = 2.

**a.** Find the copy's perimeter. *1 pt*

**b.** Find the copy's area. *1 pt*

**5.** A figure has area 6 $cm^2$. Its scale copy has area 150 $cm^2$. What is the scale factor? *1 pt*

**6.** A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

**a.** Find the real length of the wall drawn 2 in long. *1 pt*

**b.** Find the room's real area. *1 pt*

**7.** A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

**8.** A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

## Converting Across Systems *8 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 7 in? *1 pt*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 110 cm. *1 pt*

- ☐ A. 110 × 2.54
- ☐ B. 110 × 0.39
- ☐ C. 110 ÷ 0.39
- ☐ D. 110 ÷ 2.54

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 1.5 yd? *1 pt*

**12.** 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

**13.** 1 gal = 3.785 L. How many liters is 3 gal? *1 pt*

**14.** 1 in = 2.54 cm. A tile is 10 $in^2$ in area. Which is closest to its area in square centimeters? *1 pt*

- ○ A. 25.4 $cm^2$
- ○ B. 64.5 $cm^2$
- ○ C. 6.45 $cm^2$
- ○ D. 50.8 $cm^2$

**15.** Rate on Nov 5: 1 US dollar = 1.32 Canadian dollars.

**a.** How many Canadian dollars is $90? *1 pt*

**b.** How many US dollars is 66 Canadian dollars? *1 pt*

## Multi-Step Proportion Problems

*6 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 27 miles per gallon and its tank holds 10 gal.

**a.** How far does it go on a full tank? *1 pt*

**b.** It makes a 1,200-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

**17.** A recipe for 5 servings uses 2 cups of rice.

**a.** How much rice for 15 servings? *1 pt*

**b.** With 10 cups on hand, how many servings can be made? *1 pt*

**18.** Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second color. How many gallons is the whole batch? *1 pt*

**19.** 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

# Multi-Step Proportional Relationships

*24 points* **ANSWER KEY**

Name
Date
Period

*Show your work. One point for each lettered part.*

## Scale Drawings and Scale Factors

*10 points MA.7.GR.1.5*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

***k = 3***

*copy ÷ original*

*Scoring: 1 point.*

**2.** A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

***k = 0.5***

*copy ÷ original; the copy is smaller, so k is under 1 — 2 is the proportion backwards*

*Scoring: 1 point.*

**3.** A figure is scaled by a scale factor of 2. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. The area is 2 times as large.
- ☒ **B. Each side is 2 times as long.**
- ☒ **C. The perimeter is 2 times as long.**
- ☒ **D. The area is 4 times as large.**
- ☐ E. Each angle is 2 times as large.

***B, C and D***

*All three and nothing else for the point: sides and perimeter × k, area × $k^2$, angles unchanged.*

*Scoring: 1 point.*

**4.** A 3 cm by 5 cm rectangle is scaled by k = 2.

**a.** Find the copy's perimeter. *1 pt*

***32 cm***

*perimeter × k*

**b.** Find the copy's area. *1 pt*

***60 cm²***

*area × k² — 30 multiplied the area by k once*

*Scoring: a 1 · b 1 = 2 points.*

**5.** A figure has area 6 cm². Its scale copy has area 150 cm². What is the scale factor? *1 pt*

***k = 5***

*150 ÷ 6 = 25 is k², so k = √25. A board with 25 handed in the area ratio as k.*

*Scoring: 1 point.*

**6.** A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

**a.** Find the real length of the wall drawn 2 in long. *1 pt*

***10 ft***

*the scale per 1 inch, then ×: 1 in : 5 ft*

**b.** Find the room's real area. *1 pt*

***150 ft²***

*real dimensions first: 10 by 15 — 6 in² × 5 = 30 is the named error*

*Scoring: a 1 · b 1 = 2 points.*

**7.** A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

***5 in***

*real ÷ rate*

*Scoring: 1 point.*

**8.** A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

***135 ft²***

*9 ft by 15 ft.*

*Scoring: 1 point.*

## Converting Across Systems

*8 points* *MA.7.AR.3.3*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 7 in? *1 pt*

***17.78 cm (17.9 by ÷ 0.39)***

*× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it*

*Scoring: 1 point.*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 110 cm. *1 pt*

- ☐ A. 110 × 2.54
- ☒ **B. 110 × 0.39**
- ☐ C. 110 ÷ 0.39
- ☒ **D. 110 ÷ 2.54**

***B and D***

*Both sheet routes (42.9 and 43.3) and nothing else for the point.*

*Scoring: 1 point.*

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 1.5 yd? *1 pt*

***137.16 cm***

*a chain: 54 in, then × 2.54 (54 ÷ 0.39 = 138.5 by the other ratio — full credit). The estimate catches 3.81.*

*Scoring: 1 point.*

**12.** 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

***77.14 lb (77.09 by ÷ 0.454)***

*× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**13.** 1 gal = 3.785 L. How many liters is 3 gal? *1 pt*

***11.355 L (11.36 by ÷ 0.264)***

*more liters than gallons; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**14.** 1 in = 2.54 cm. A tile is 10 $in^2$ in area. Which is closest to its area in square centimeters? *1 pt*

- ○ A. 25.4 $cm^2$
- ● **B. 64.5 $cm^2$**
- ○ C. 6.45 $cm^2$
- ○ D. 50.8 $cm^2$

***B***

*$10 \times 2.54^2 = 64.52$ (or $10 \div 0.39^2 = 65.7$ — the sheet's two routes differ here; either picks B).*

*Scoring: 1 point.*

**15.** Rate on Nov 5: 1 US dollar = 1.32 Canadian dollars.

**a.** How many Canadian dollars is $90? *1 pt*

***118.80 Canadian dollars***

*× the rate*

**b.** How many US dollars is 66 Canadian dollars? *1 pt*

***$50***

*÷ the rate; 87.12 multiplied both ways*

*Scoring: a 1 · b 1 = 2 points.*

## Multi-Step Proportion Problems

*6 points MA.7.AR.4.5*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 27 miles per gallon and its tank holds 10 gal.

**a.** How far does it go on a full tank? *1 pt*

***270 mi***

*the rate × the gallons*

**b.** It makes a 1,200-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

***5 tanks (1,200 ÷ 270 ≈ 4.44)***

*the bracket on the paper settles the rounding; follow-through on part (a)*

*Scoring: a 1 · b 1 = 2 points.*

**17.** A recipe for 5 servings uses 2 cups of rice.

**a.** How much rice for 15 servings? *1 pt*

***6 cups***

*k = 3 from the yields, then the ingredient*

**b.** With 10 cups on hand, how many servings can be made? *1 pt*

***25 servings***

*5 recipes × the yield; 50 scaled the yield by the cups*

*Scoring: a 1 · b 1 = 2 points.*

**18.** Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second color. How many gallons is the whole batch? *1 pt*

***24 gal***

*k = 4 from the second color, then the whole, 6 parts; 12 found the first color and stopped*

*Scoring: 1 point.*

**19.** 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

***The train: 45 mi/h = 72.45 km/h, and 80 > 72.45.***

*Same unit first; 80 > 45 compared two units.*

*Scoring: 1 point.*

## Score Tracker

| Benchmark | Questions | Possible | Earned |
|---|---|---|---|
| MA.7.GR.1.5 | 1, 2, 3, 4, 5, 6, 7, 8 | 10 | |
| MA.7.AR.3.3 | 9, 10, 11, 12, 13, 14, 15 | 8 | |
| MA.7.AR.4.5 | 16, 17, 18, 19 | 6 | |
| **Total** | | **24** | |

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.*

GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

# Multi-Step Proportional Relationships

*24 points* **WORKED ANSWERS**

*Show your work. One point for each lettered part.*

## Scale Drawings and Scale Factors *10 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

***k = 3***

*copy ÷ original*

*Scoring: 1 point.*

**2.** A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

***k = 0.5***

*copy ÷ original; the copy is smaller, so k is under 1 — 2 is the proportion backwards*

*Scoring: 1 point.*

**3.** A figure is scaled by a scale factor of 2. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. The area is 2 times as large.
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- ☒ **C. The perimeter is 2 times as long.**
- ☒ **D. The area is 4 times as large.**
- ☐ E. Each angle is 2 times as large.

***B, C and D***

*All three and nothing else for the point: sides and perimeter × k, area × $k^2$, angles unchanged.*

*Scoring: 1 point.*

**4.** A 3 cm by 5 cm rectangle is scaled by k = 2.

**a.** Find the copy's perimeter. *1 pt*

***32 cm***

*perimeter × k*

**b.** Find the copy's area. *1 pt*

***60 cm²***

*area × k² — 30 multiplied the area by k once*

*Scoring: a 1 · b 1 = 2 points.*

**5.** A figure has area 6 cm². Its scale copy has area 150 cm². What is the scale factor? *1 pt*

***k = 5***

*150 ÷ 6 = 25 is k², so k = √25. A board with 25 handed in the area ratio as k.*

*Scoring: 1 point.*

**6.** A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

**a.** Find the real length of the wall drawn 2 in long. *1 pt*

***10 ft***

*the scale per 1 inch, then ×: 1 in : 5 ft*

**b.** Find the room's real area. *1 pt*

***150 ft²***

*real dimensions first: 10 by 15 — 6 in² × 5 = 30 is the named error*

*Scoring: a 1 · b 1 = 2 points.*

**7.** A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

***5 in***

*real ÷ rate*

*Scoring: 1 point.*

**8.** A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

***135 ft²***

*9 ft by 15 ft.*

*Scoring: 1 point.*

## Converting Across Systems

*8 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 7 in? *1 pt*

***17.78 cm (17.9 by ÷ 0.39)***

*× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it*

*Scoring: 1 point.*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 110 cm. *1 pt*

- ☐ A. 110 × 2.54
- ☒ **B. 110 × 0.39**
- ☐ C. 110 ÷ 0.39
- ☒ **D. 110 ÷ 2.54**

***B and D***

*Both sheet routes (42.9 and 43.3) and nothing else for the point.*

*Scoring: 1 point.*

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 1.5 yd? *1 pt*

***137.16 cm***

*a chain: 54 in, then × 2.54 (54 ÷ 0.39 = 138.5 by the other ratio — full credit). The estimate catches 3.81.*

*Scoring: 1 point.*

**12.** 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

***77.14 lb (77.09 by ÷ 0.454)***

*× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**13.** 1 gal = 3.785 L. How many liters is 3 gal? *1 pt*

***11.355 L (11.36 by ÷ 0.264)***

*more liters than gallons; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**14.** 1 in = 2.54 cm. A tile is 10 $in^2$ in area. Which is closest to its area in square centimeters? *1 pt*

- ○ A. 25.4 $cm^2$
- ● **B. 64.5 $cm^2$**
- ○ C. 6.45 $cm^2$
- ○ D. 50.8 $cm^2$

***B***

*$10 \times 2.54^2 = 64.52$ (or $10 \div 0.39^2 = 65.7$ — the sheet's two routes differ here; either picks B).*

*Scoring: 1 point.*

**15.** Rate on Nov 5: 1 US dollar = 1.32 Canadian dollars.

**a.** How many Canadian dollars is $90? *1 pt*

***118.80 Canadian dollars***

*× the rate*

**b.** How many US dollars is 66 Canadian dollars? *1 pt*

***$50***

*÷ the rate; 87.12 multiplied both ways*

*Scoring: a 1 · b 1 = 2 points.*

## Multi-Step Proportion Problems *6 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 27 miles per gallon and its tank holds 10 gal.

**a.** How far does it go on a full tank? *1 pt*

***270 mi***

*the rate × the gallons*

**b.** It makes a 1,200-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

***5 tanks (1,200 ÷ 270 ≈ 4.44)***

*the bracket on the paper settles the rounding; follow-through on part (a)*

*Scoring: a 1 · b 1 = 2 points.*

**17.** A recipe for 5 servings uses 2 cups of rice.

**a.** How much rice for 15 servings? *1 pt*

***6 cups***

*k = 3 from the yields, then the ingredient*

**b.** With 10 cups on hand, how many servings can be made? *1 pt*

***25 servings***

*5 recipes × the yield; 50 scaled the yield by the cups*

*Scoring: a 1 · b 1 = 2 points.*

**18.** Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second color. How many gallons is the whole batch? *1 pt*

***24 gal***

*k = 4 from the second color, then the whole, 6 parts; 12 found the first color and stopped*

*Scoring: 1 point.*

**19.** 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

***The train: 45 mi/h = 72.45 km/h, and 80 > 72.45.***

*Same unit first; 80 > 45 compared two units.*

*Scoring: 1 point.*

GRADE 7 ACCELERATED · UNIT 5 · LESSON 1

# Scale Drawings

*Independent Set 5.01* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** A 5 cm side becomes 15 cm in the copy. Find the scale factor.

***k = 3***

**2.** A 15 cm side becomes 6 cm in the copy. Find the scale factor.

***k = 0.4***

*The copy is smaller, so k is less than 1.*

**3.** A 3 cm by 5 cm rectangle is copied at 7.5 cm by 12.5 cm. Find the scale factor.

***k = 2.5***

*7.5 ÷ 3 and 12.5 ÷ 5 both give 2.5.*

**4.** The scale factor is 1.5 and a side of the original is 8 cm. How long is that side in the copy?

***12 cm***

**5.** A map's scale is $1\text{ in} : 25\text{ mi}$. Two cities are 2.8 in apart on the map. How far apart are they?

***70 mi***

**6.** On the same map, a road is 150 mi long. How long is it on the map?

***6 in***

*Back to the map: divide by 25.*

GRADE 7 ACCELERATED · UNIT 5 · LESSON 1

# Scale Drawings

*Question Bank 5.01* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, find the scale factor from the original to the copy.**

**1.** A 4 cm side becomes 12 cm.

***k = 3***

**2.** A 9 cm side becomes 3 cm.

***k = ⅓***

**3.** A 2 cm by 7 cm rectangle is copied at 5 cm by 17.5 cm.

***k = 2.5***

*5 ÷ 2 and 17.5 ÷ 7 both give 2.5.*

**4.** A 16 cm side becomes 12 cm.

***k = ¾***

**For questions 5 through 8, use the scale factor or the scale.**

**5.** The scale factor is 4 and a side of the original is 2.5 cm. How long is that side in the copy?

***10 cm***

**6.** The scale factor is 0.5 and a side of the original is 9 cm. How long is that side in the copy?

***4.5 cm***

**7.** A map's scale is $1\,\text{cm} : 30\,\text{km}$ . Two towns are 5.5 cm apart on the map. How far apart are they?

***165 km***

**8.** On the same map, a lake is 45 km long. How long is it on the map?

***1.5 cm***

*Back to the map: divide by 30.*

**Choose one.**

**9.** A drawing has a 65° angle. The real figure is a scale copy at $k = 3$. What is the angle in the real figure?

- ○ A. 68°
- ○ B. 195°
- ● **C. 65°**
- ○ D. 21.7°

***C***

*A scale factor multiplies lengths; angles do not change.*

GRADE 7 ACCELERATED · UNIT 5 · LESSON 1

# Scale Drawings

*Question Bank - Additional 5.01* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, find the scale factor from the original to the copy.**

**1.** A 3 cm side becomes 12 cm.

***k = 4***

**2.** A 10 cm side becomes 2 cm.

***k =*** $\frac{1}{5}$

**3.** A 3 cm by 4 cm rectangle is copied at 4.5 cm by 6 cm.

***k = 1.5***

*4.5 ÷ 3 and 6 ÷ 4 both give 1.5.*

**4.** A 20 cm side becomes 15 cm.

***k = ¾***

**For questions 5 through 8, use the scale factor or the scale.**

**5.** The scale factor is 3 and a side of the original is 3.5 cm. How long is that side in the copy?

***10.5 cm***

**6.** The scale factor is 0.25 and a side of the original is 12 cm. How long is that side in the copy?

***3 cm***

**7.** A map's scale is $1\text{ in} : 40\text{ mi}$ . Two cities are 3.25 in apart on the map. How far apart are they?

***130 mi***

**8.** On the same map, a river is 90 mi long. How long is it on the map?

***2.25 in***

*Back to the map: divide by 40.*

**Choose one.**

**9.** A drawing has a 30° angle. The real figure is a scale copy at $k = 4$. What is the angle in the real figure?

- ○ A. 34°
- ○ B. 120°
- ○ C. 7.5°
- ● **D. 30°**

***D***

*A scale factor multiplies lengths; angles do not change.*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

# Scale Factor: Perimeter and Area

*Independent Set 5.02* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** A figure with perimeter 18 cm is scaled by $k = 3$. Find the copy's perimeter.

***54 cm***

**2.** A figure with area $7\,\text{cm}^2$ is scaled by $k = 3$. Find the copy's area.

***63 cm²***

*Area × $k^2$ = 7 × 9.*

**3.** A figure with area $40\,\text{cm}^2$ is scaled by $k = \frac{1}{2}$. Find the copy's area.

***10 cm²***

*Area × (½)² = 40 × ¼.*

**4.** A 2 cm by 5 cm rectangle is scaled by $k = 2.5$. Find the copy's perimeter and area.

***P = 35 cm, A = 62.5 cm²***

*P = 14 × 2.5; A = 10 × 6.25.*

**5.** A figure has area $9\,\text{cm}^2$ and its scale copy has area $144\,\text{cm}^2$. Find $k$.

***k = 4***

*144 ÷ 9 = 16 = $k^2$, so k = √16.*

**6.** A figure's perimeter is 12 cm and its copy's perimeter is 48 cm. What is the area ratio, copy to original?

***16***

*k = 4 from the perimeters; the area ratio is $k^2$ = 16.*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

# Scale Factor: Perimeter and Area

*Question Bank 5.02* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, find the copy's perimeter or area.**

**1.** Perimeter 22 cm, $k = 2$ .

***44 cm***

**2.** Area $5\,\text{cm}^2$ , $k = 4$ .

***80 cm²***

*5 × 16.*

**3.** Area $72\,\text{cm}^2$ , $k = \frac{1}{3}$ .

***8 cm²***

*72 ×* $\frac{1}{9}$ .

**4.** A 3 cm by 7 cm rectangle, $k = 2$ . Perimeter and area of the copy.

***P = 40 cm, A = 84 cm²***

*20 × 2; 21 × 4.*

**For questions 5 through 7, find $k$ , or the ratio asked.**

**5.** Areas $6\,\text{cm}^2$ and $150\,\text{cm}^2$ . Find $k$ .

***k = 5***

*150 ÷ 6 = 25 = $k^2$.*

**6.** Areas $50\,\text{cm}^2$ and $8\,\text{cm}^2$ , original to copy. Find $k$ .

***k = 0.4***

*8 ÷ 50 = 0.16 = $k^2$, so k = 0.4: the copy is smaller.*

**7.** Perimeters 15 cm and 45 cm, original to copy. What is the area ratio?

***9***

*k = 3; $k^2$ = 9.*

**Choose one.**

**8.** A figure with area $6\,\text{cm}^2$ is scaled by $k = 5$. What is the copy's area?

- ○ A. $30\,\text{cm}^2$
- ○ B. $60\,\text{cm}^2$
- ● **C.** $150\,\text{cm}^2$
- ○ D. $11\,\text{cm}^2$

***C***

*$6 \times 5^2 = 6 \times 25$.*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

# Scale Factor: Perimeter and Area

*Question Bank - Additional 5.02* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, find the copy's perimeter or area.**

**1.** Perimeter 26 cm, $k = 3$ .

***78 cm***

**2.** Area $9\,\text{cm}^2$ , $k = 3$ .

***81 cm²***

*9 × 9.*

**3.** Area $100\,\text{cm}^2$ , $k = \frac{1}{2}$ .

***25 cm²***

*100 × ¼.*

**4.** A 4 cm by 5 cm rectangle, $k = 1.5$ . Perimeter and area of the copy.

***P = 27 cm, A = 45 cm²***

*18 × 1.5; 20 × 2.25.*

**For questions 5 through 7, find $k$ , or the ratio asked.**

**5.** Areas $11\,\text{cm}^2$ and $99\,\text{cm}^2$ . Find $k$ .

***k = 3***

*99 ÷ 11 = 9 = $k^2$.*

**6.** Areas $32\,\text{cm}^2$ and $2\,\text{cm}^2$ , original to copy. Find $k$ .

***k = ¼***

*2 ÷ 32 = 1⁄16 = $k^2$, so k = ¼.*

**7.** Perimeters 10 cm and 50 cm, original to copy. What is the area ratio?

***25***

*k = 5; $k^2$ = 25.*

**Choose one.**

**8.** A figure with area $4\,\text{cm}^2$ is scaled by $k = 6$. What is the copy's area?

- ○ A. $48\,\text{cm}^2$
- ● **B.** $144\,\text{cm}^2$
- ○ C. $24\,\text{cm}^2$
- ○ D. $10\,\text{cm}^2$

***B***

*$4 \times 6^2 = 4 \times 36$.*

GRADE 7 ACCELERATED · UNIT 5 · LESSON 4

# Real-World Scale Drawings

*Independent Set 5.04* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** A plan is drawn at $\frac{1}{2}\text{ in} : 1\text{ ft}$. A wall is 6 in long on it. Find the real length.

***12 ft***

*1 in : 2 ft; 6 × 2.*

**2.** A plan is drawn at $1\text{ in} : 5\text{ ft}$. A room is 3.2 in long on it. Find the real length.

***16 ft***

**3.** A plan is drawn at $1\text{ in} : 5\text{ ft}$. A real wall is 22 ft long. Find its length on the plan.

***4.4 in***

*22 ÷ 5.*

**4.** A plan is drawn at $\frac{1}{4}\text{ in} : 1\text{ ft}$. A room is 2 in by 3.5 in on it. Find the real dimensions and area.

***8 ft by 14 ft; 112 ft$^2$***

*1 in : 4 ft; 2 × 4 and 3.5 × 4; then 8 × 14.*

**5.** A model boat is built at $1 : 40$ and is 15 cm long. Find the real boat's length in meters.

***6 m***

*15 × 40 = 600 cm.*

**6.** On a drawing a wall is 3 in long; the real wall is 16.5 ft. Another wall is 5 in long on the drawing. How long is it really?

***27.5 ft***

*The pair gives 5.5 ft per inch; 5 × 5.5.*

GRADE 7 ACCELERATED · UNIT 5 · LESSON 4

# Real-World Scale Drawings

*Question Bank 5.04* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, use the scale given.**

**1.** $\frac{1}{2}$ in : 1 ft . A wall is 5 in on the plan. Real length?

***10 ft***

*1 in : 2 ft.*

**2.** 1 in : 6 ft . A room is 2.5 in on the plan. Real length?

***15 ft***

**3.** 1 in : 6 ft . A real wall is 27 ft. Length on the plan?

***4.5 in***

*27 ÷ 6.*

**4.** $\frac{1}{4}$ in : 1 ft . A room is 1.5 in by 2.5 in on the plan. Real dimensions and area?

***6 ft by 10 ft; 60 ft²***

*1 in : 4 ft; 6 × 10.*

**For questions 5 through 7, answer in the unit asked.**

**5.** A model train is built at 1 : 80 and is 25 cm long. The real train, in meters?

***20 m***

*25 × 80 = 2,000 cm.*

**6.** On a drawing a wall is 2 in long; the real wall is 15 ft. Another wall is 7 in long on the drawing. How long is it really?

***52.5 ft***

*7.5 ft per inch; 7 × 7.5.*

**7.** A 60-ft boat is modeled at 1 : 60 and at 1 : 90 . Which model is longer, and by how much?

***The 1 : 60 model, by 4 in (12 in against 8 in)***

*60 ft = 720 in; 720 ÷ 60 = 12; 720 ÷ 90 = 8.*

**Choose one.**

**8.** A plan is drawn at $1\text{ in} : 3\text{ ft}$ . A room is 2 in by 4 in on it. What is the room's real area?

- ○ A. $18\,\text{ft}^2$
- ○ B. $36\,\text{ft}^2$
- ○ C. $24\,\text{ft}^2$
- ● **D.** $72\,\text{ft}^2$

***D***

*6 ft by 12 ft; 6 × 12.*

GRADE 7 ACCELERATED · UNIT 5 · LESSON 4

# Real-World Scale Drawings

*Question Bank - Additional 5.04* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, use the scale given.**

**1.** $\frac{1}{2}$ in : 1 ft . A wall is 7 in on the plan. Real length?

***14 ft***

*1 in : 2 ft.*

**2.** 1 in : 8 ft . A room is 1.5 in on the plan. Real length?

***12 ft***

**3.** 1 in : 8 ft . A real wall is 36 ft. Length on the plan?

***4.5 in***

*36 ÷ 8.*

**4.** $\frac{1}{4}$ in : 1 ft . A room is 2.5 in by 3.5 in on the plan. Real dimensions and area?

***10 ft by 14 ft; 140 ft$^2$***

*1 in : 4 ft; 10 × 14.*

**For questions 5 through 7, answer in the unit asked.**

**5.** A model plane is built at 1 : 60 and is 30 cm long. The real plane, in meters?

***18 m***

*30 × 60 = 1,800 cm.*

**6.** On a drawing a wall is 4 in long; the real wall is 22 ft. Another wall is 7 in long on the drawing. How long is it really?

***38.5 ft***

*5.5 ft per inch; 7 × 5.5.*

**7.** A 36-ft bus is modeled at 1 : 36 and at 1 : 54 . Which model is longer, and by how much?

***The 1 : 36 model, by 4 in (12 in against 8 in)***

*36 ft = 432 in; 432 ÷ 36 = 12; 432 ÷ 54 = 8.*

**Choose one.**

**8.** A plan is drawn at $1\text{ in} : 5\text{ ft}$ . A room is 2 in by 3 in on it. What is the room's real area?

- ○ A. $25\,\text{ft}^2$
- ○ B. $30\,\text{ft}^2$
- ● **C.** $150\,\text{ft}^2$
- ○ D. $50\,\text{ft}^2$

***C***

*10 ft by 15 ft; 10 × 15.*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 6–7

# Converting Length Across Systems; Currency

*Independent Set 5.06* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** $1\text{ in} = 2.54\text{ cm}$ . How many centimeters is 12 in?

***30.48 cm***

**2.** $1\text{ cm} = 0.39\text{ in}$ . How many inches is 40 cm?

***15.6 in***

*40 × 0.39 (or 40 ÷ 2.54 = 15.7).*

**3.** $1\text{ mi} = 1.61\text{ km}$ . How many kilometers is 10 mi?

***16.1 km***

**4.** $1\text{ km} = 0.62\text{ mi}$ . How many miles is 8 km?

***4.96 mi***

*8 × 0.62.*

**5.** $1\text{ yd} = 3\text{ ft}$ , $1\text{ ft} = 12\text{ in}$ , $1\text{ in} = 2.54\text{ cm}$ . How many centimeters is 2 yd? Estimate first.

***About 180; exactly 182.88 cm***

*2 yd = 72 in; 72 × 2.54.*

**6.** Today's rate: $1\text{ US dollar} = 0.92\text{ euros}$ . How many euros is $50? How many dollars is 23 euros?

***46 euros; $25***

*50 × 0.92; 23 ÷ 0.92.*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 6–7

# Converting Length Across Systems; Currency

*Question Bank 5.06* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, convert with the ratio given.**

**1.** $1\text{ in} = 2.54\text{ cm}$ . 8 in to centimeters.

***20.32 cm***

**2.** $1\text{ cm} = 0.39\text{ in}$ . 60 cm to inches.

***23.4 in***

*60 × 0.39 (or 60 ÷ 2.54 = 23.6).*

**3.** $1\text{ mi} = 1.61\text{ km}$ . 20 mi to kilometers.

***32.2 km***

**4.** $1\text{ km} = 0.62\text{ mi}$ . 15 km to miles.

***9.3 mi***

*15 × 0.62.*

**For questions 5 through 7, show every step.**

**5.** $1\text{ ft} = 12\text{ in}$ , $1\text{ in} = 2.54\text{ cm}$ . 5 ft to centimeters.

***152.4 cm***

*60 in × 2.54.*

**6.** Today's rate: $1\text{ US dollar} = 1.35\text{ Canadian dollars}$ . $40 to Canadian dollars; 27 Canadian dollars to US dollars.

***54 CAD; $20***

*40 × 1.35; 27 ÷ 1.35.*

**7.** Rates today: $1\text{ US dollar} = 1.32\text{ Canadian dollars}$ ; $1\text{ Canadian dollar} = 16.2\text{ pesos}$ . $50 to pesos, by way of Canadian dollars.

***1,069.20 pesos***

*50 × 1.32 = 66 CAD; 66 × 16.2.*

**Choose one.**

**8.** $1\,\text{km} = 0.62\,\text{mi}$ . A road is 20 km long. How many miles is that?

- ● **A. 12.4 mi**
- ○ B. 32.2 mi
- ○ C. 1.24 mi
- ○ D. 20.62 mi

***A***

*20 × 0.62.*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 6–7

# Converting Length Across Systems; Currency

*Question Bank - Additional 5.06* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, convert with the ratio given.**

**1.** $1\text{ in} = 2.54\text{ cm}$ . 9 in to centimeters.

***22.86 cm***

**2.** $1\text{ cm} = 0.39\text{ in}$ . 80 cm to inches.

***31.2 in***

*80 × 0.39 (or 80 ÷ 2.54 = 31.5).*

**3.** $1\text{ mi} = 1.61\text{ km}$ . 30 mi to kilometers.

***48.3 km***

**4.** $1\text{ km} = 0.62\text{ mi}$ . 25 km to miles.

***15.5 mi***

*25 × 0.62.*

**For questions 5 through 7, show every step.**

**5.** $1\text{ ft} = 12\text{ in}$ , $1\text{ in} = 2.54\text{ cm}$ . 4 ft to centimeters.

***121.92 cm***

*48 in × 2.54.*

**6.** Today's rate: $1\text{ US dollar} = 0.92\text{ euros}$ . $30 to euros; 69 euros to US dollars.

***27.60 euros; $75***

*30 × 0.92; 69 ÷ 0.92.*

**7.** Rates today: $1\text{ US dollar} = 1.32\text{ Canadian dollars}$ ; $1\text{ Canadian dollar} = 16.2\text{ pesos}$ . $25 to pesos, by way of Canadian dollars.

***534.60 pesos***

*25 × 1.32 = 33 CAD; 33 × 16.2.*

**Choose one.**

**8.** $1\,\text{mi} = 1.61\,\text{km}$ . A road is 40 mi long. How many kilometers is that?

- ○ A. 24.8 km
- ○ B. 6.44 km
- ● **C. 64.4 km**
- ○ D. 41.61 km

***C***

*40 × 1.61.*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 8–9

# Converting Mass, Area and Volume

*Independent Set 5.08* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** $1\,\text{lb} = 0.454\,\text{kg}$ . How many kilograms is 30 lb?

***13.62 kg***

*(30 ÷ 2.204 = 13.61 by the other ratio.)*

**2.** $1\,\text{kg} = 2.204\,\text{lb}$ . How many pounds is 25 kg?

***55.1 lb***

**3.** $1\,\text{gal} = 3.785\,\text{L}$ . How many liters is 6 gal?

***22.71 L***

**4.** $1\,\text{ft} = 12\,\text{in}$ . How many square inches is $9\,\text{ft}^2$ ?

***1,296 in²***

*9 × 144: the ratio squared.*

**5.** $1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$ . How many liters is 24 fl oz?

***0.72 L***

*3 c × 0.24.*

**6.** $1\,\text{ft} = 0.305\,\text{m}$ . How many square meters is $30\,\text{ft}^2$ ? (Round to two places.)

***About 2.79 m²***

*30 × 0.305²: the sheet's length ratio, squared, across systems.*

# Converting Mass, Area and Volume

*Question Bank 5.08* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, convert with the ratio given.**

**1.** $1\,\text{lb} = 0.454\,\text{kg}$ . 20 lb to kilograms.

***9.08 kg***

*(20 ÷ 2.204 = 9.07 by the other ratio.)*

**2.** $1\,\text{kg} = 2.204\,\text{lb}$ . 15 kg to pounds.

***33.06 lb***

**3.** $1\,\text{gal} = 3.785\,\text{L}$ . 8 gal to liters.

***30.28 L***

**4.** $1\,\text{L} = 0.264\,\text{gal}$ . 25 L to gallons.

***6.6 gal***

**For questions 5 through 7, say how many times the ratio is used.**

**5.** $1\,\text{yd} = 3\,\text{ft}$ . $4\,\text{yd}^2$ to square feet.

***36 ft²***

*4 × 9: twice.*

**6.** $1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$ . 32 fl oz to liters.

***0.96 L***

*4 c × 0.24.*

**7.** $1\,\text{ft} = 12\,\text{in}$ . $3\,\text{ft}^3$ to cubic inches.

***5,184 in³***

*3 × 1,728: three times.*

**Choose one.**

**8.** $1\,\text{ft} = 12\,\text{in}$ . A rug is $6\,\text{ft}^2$ in area. How many square inches is that?

- ● **A.** $864\,\text{in}^2$
- ○ B. $72\,\text{in}^2$
- ○ C. $144\,\text{in}^2$
- ○ D. $150\,\text{in}^2$

***A***

*6 × 144.*

# Converting Mass, Area and Volume

*Question Bank - Additional 5.08* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, convert with the ratio given.**

**1.** $1\text{ lb} = 0.454\text{ kg}$ . 40 lb to kilograms.

***18.16 kg***

*(40 ÷ 2.204 = 18.15 by the other ratio.)*

**2.** $1\text{ kg} = 2.204\text{ lb}$ . 5 kg to pounds.

***11.02 lb***

**3.** $1\text{ gal} = 3.785\text{ L}$ . 12 gal to liters.

***45.42 L***

**4.** $1\text{ L} = 0.264\text{ gal}$ . 50 L to gallons.

***13.2 gal***

**For questions 5 through 7, say how many times the ratio is used.**

**5.** $1\text{ yd} = 3\text{ ft}$ . $7\text{ yd}^2$ to square feet.

***63 ft²***

*7 × 9: twice.*

**6.** $1\text{ c} = 8\text{ fl oz}$ and $1\text{ c} = 0.24\text{ L}$ . 40 fl oz to liters.

***1.2 L***

*5 c × 0.24.*

**7.** $1\text{ ft} = 12\text{ in}$ . $4\text{ ft}^3$ to cubic inches.

***6,912 in³***

*4 × 1,728: three times.*

**Choose one.**

**8.** $1\,\text{m} = 100\,\text{cm}$ . A floor is $3\,\text{m}^2$ in area. How many square centimeters is that?

- ○ A. $600\,\text{cm}^2$
- ○ B. $10{,}000\,\text{cm}^2$
- ○ C. $300\,\text{cm}^2$
- ● **D.** $30{,}000\,\text{cm}^2$

***D***

*3 × 10,000.*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 10–11

# Multi-Step Proportion Problems

*Independent Set 5.10* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** A car gets 32 miles per gallon and its tank holds 14 gal. How far does it go on a tank?

***448 mi***

**2.** That car makes a 1,000-mi trip. How many tanks of gas does it use? (Part of a tank counts as a whole tank.)

***3 tanks***

*1,000 ÷ 448 ≈ 2.23 → 3.*

**3.** A recipe for 4 servings uses $1\frac{1}{4}$ cups of rice. How much rice for 10 servings?

***3⅛ cups***

*k = 2.5; 2.5 × 1¼.*

**4.** $1\,\text{km} = 0.62\,\text{mi}$. A train does 120 km/h; a bus does 70 mi/h. Which is faster? Show the speeds in one unit.

***The train: 120 km/h = 74.4 mi/h > 70 mi/h***

*120 × 0.62.*

**5.** Juice is mixed orange : mango : water = $3:2:5$. A batch uses 9 cups of orange. How many cups is the whole batch?

***30 cups***

*k = 3; 10 parts × 3.*

**6.** A plan is drawn at $1\,\text{in} : 6\,\text{ft}$; a room is 2 in by 3.5 in on it. Find the room's real area.

***252 ft²***

*12 ft by 21 ft. (The square-meter version is the test's transfer item and is not practised here — ruling 18.)*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 10–11

# Multi-Step Proportion Problems

*Question Bank 5.10* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 3, one rate, used as many times as it takes.**

**1.** A car gets 25 miles per gallon with a 16-gal tank. Miles on a full tank?

***400 mi***

**2.** That car makes a 1,500-mi trip. How many tanks of gas does it use? (Part of a tank counts as a whole tank.)

***4 tanks***

*1,500 ÷ 400 = 3.75 → 4.*

**3.** A printer prints 42 pages per minute. How long does a 1,260-page job take?

***30 min***

**For questions 4 through 6, the same unit first.**

**4.** $1\,\text{mi} = 1.61\,\text{km}$. Runner A: 3 mi in 24 min. Runner B: 5 km in 25 min. Each speed in km/h, and who is faster?

***A ≈ 12.08 km/h, B = 12 km/h; A***

*4.83 ÷ 0.4; 5 ÷ (5⁄12).*

**5.** A recipe for 6 servings uses $2\frac{1}{4}$ cups of flour. Flour for 16 servings? With 9 cups on hand, how many servings?

***6 cups; 24 servings***

*k = 8⁄3; 9 ÷ 2¼ = 4 recipes × 6.*

**6.** Paint is mixed $1:3:4$. A batch uses 6 gal of the second color. The whole batch?

***16 gal***

*k = 2; 8 parts × 2.*

**Choose one.**

**7.** A car gets 30 miles per gallon with a 10-gal tank. It makes a 750-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.)

- ● **A. 3**
- ○ B. 25
- ○ C. 75
- ○ D. 2

***A***

*300 mi per tank; 750 ÷ 300 = 2.5 → 3.*

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 10–11

# Multi-Step Proportion Problems

*Question Bank - Additional 5.10* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 3, one rate, used as many times as it takes.**

**1.** A car gets 24 miles per gallon with a 15-gal tank. Miles on a full tank?

***360 mi***

**2.** That car makes a 1,300-mi trip. How many tanks of gas does it use? (Part of a tank counts as a whole tank.)

***4 tanks***

*1,300 ÷ 360 ≈ 3.6 → 4.*

**3.** A printer prints 36 pages per minute. How long does a 900-page job take?

***25 min***

**For questions 4 through 6, the same unit first.**

**4.** $1\,\text{km} = 0.62\,\text{mi}$ . Car A: 90 km/h. Car B: 60 mi/h. Which is faster? Show both in mi/h.

***B: 90 km/h = 55.8 mi/h < 60***

*90 × 0.62.*

**5.** A recipe for 8 servings uses 3 cups of flour. Flour for 12 servings? With 9 cups on hand, how many servings?

***4½ cups; 24 servings***

*k = 1.5; 9 ÷ 3 = 3 recipes × 8.*

**6.** Paint is mixed $2:3:5$ . A batch uses 15 gal of the third color. The whole batch?

***30 gal***

*k = 3; 10 parts × 3.*

**Choose one.**

**7.** A car gets 35 miles per gallon with a 12-gal tank. It makes a 1,050-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.)

- ○ A. 30
- ● **B. 3**
- ○ C. 2
- ○ D. 87.5

***B***

*420 mi per tank; 1,050 ÷ 420 = 2.5 → 3.*